A CURE FOR EVERYONE: A GUIDE TO IMPLEMENTING COURSE-BASED UNDERGRADUATE RESEARCH EXPERIENCES

Jonathan J.-M. Calède
The Ohio State University

LibreTexts™

A CURE for everyone: A guide to implementing Course-based Undergraduate Research Experiences (Calède)

The LibreTexts libraries are Powered by NICE CXone Expert and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

This text was compiled on 08/14/2026

TABLE OF CONTENTS

Licensing LicensingEdit Update titlecancel

A detailed breakdown of this resource's licensing can be found in ***Back Matter/Detailed Licensing***.

CHAPTER OVERVIEW

1: COURSE-BASED UNDERGRADUATE RESEARCH EXPERIENCES AT THE OHIO STATE UNIVERSITY

1.1: Research, Inquiry, and Scholarship in the GE at Ohio State 1.1: Research, Inquiry, and Scholarship in the GE at Ohio StateEdit Update titlecancel

A) Research, Inquiry, and Scholarship in the GE at Ohio State

Jonathan J.-M. Calède

The new general education (GE) requirements of the Ohio State University frame the experience of students at Ohio state around foundation courses, theme courses, and bookends of the four-year degree. One element of this framework explicitly offers the opportunity for instructors to weave in undergraduate research. Indeed, as part of the GE, students can take a single 4+ credit course in their sophomore or junior year (instead of two 3 credit courses in two different disciplines) to satisfy the GE Theme requirement on the condition that this course be recognized as high-impact (GE implementation committee report). One of these high-impact practices is research and creative inquiry. The GE identifies three broad learning goals for classes implementing undergraduate research (Research and creative inquiry courses description):

- Provide students with training in the tools and methodology of the discipline
- Are designed to scaffold undergraduate research and creative experiences, such that early curricular experiences provide students with the transferable skills to subsequently undertake appropriately advanced, scholarly projects
- Impress upon students the value of understanding methods and research results, noting that students undertaking scholarly work must be prepared to read and interpret primary literature

Each course that is submitted to qualify for the high-impact designation will be evaluated using an integrative practices inventory (Research and creative inquiry course inventory). This inventory articulates the following requirements of courses implementing high-impact student research experiences:

1. Performance expectations set at appropriately high levels (e.g., students investigate their own questions or develop their own creative projects)
2. Significant investment of time and effort by students over an extended period of time (e.g., scaffolded scientific or creative processes building across the term, including, e.g., reviewing literature, developing methods, collecting data, interpreting or developing a concept or idea into a full-fledged production or artistic work)
3. Interactions with faculty and peers about substantive matters including regular, meaningful faculty mentoring and peer support.
4. Students will get frequent, timely, and constructive feedback on their work, iteratively scaffolding research or creative skills in curriculum to build over time
5. Periodic, structured opportunities to reflect and integrate learning in which students interpret findings or reflect on creative work
6. Opportunities to discover relevance of learning through real-world applications (e.g., mechanism for allowing students to see their focused research question or creative project as part of a larger conceptual framework)
7. Public demonstration of competence, such as a significant public communication of research or display of creative work, or a community scholarship celebration
8. Experiences with diversity wherein students demonstrate intercultural competence and empathy with people and worldview frameworks that may differ from their own
9. Explicit and intentional efforts to promote inclusivity and a sense of belonging and safety for students, (e.g. universal design for learning principles, culturally responsible pedagogy)
10. Clear plan to market this course to get a wider enrollment of typically underserved populations.

One model of implementation of high-impact research in undergraduate classes are Course-based Undergraduate Research Experiences (CUREs).

1.2: How do CUREs satisfy GE requirements? 1.2: How do CUREs satisfy GE requirements?Edit Update titlecancel

B) How do CUREs satisfy GE requirements?

Jonathan J.-M. Calède

A CURE is a high-impact practice and represents a transformative experience for students because it:

1. Leads students through an authentic experience that reflects the work of professional scholars and meaningfully contribute to society.
2. Guides students through the creation of novel content, not merely the consumption of existing knowledge.
3. Provides the opportunity to demonstrate competence and creativity to peers and the world.
4. Fosters the development of widely applicable skills and metacognition.
5. Engages students in working collaboratively with a diverse group of peers and professionals.
6. Helps reduce the equity gap and open professional opportunities to all.

CUREs enable the implementation of research projects for an entire class. Additionally their features correspond to the expectations of high-impact research course at Ohio State articulated in the research and creative inquiry (RCI) inventory (**Figure 1**). CUREs are therefore well-suited to Theme courses with a research and creative inquiry component; they are also appropriate for other elements of the curriculum, including within the majors, that seek to bring scholarship to a greater number of undergraduate students.

Two elements (9-10) of the RCI inventory that are not included in **Figure 1** deserve attention. They are both associated with the importance of an inclusive curriculum that explicitly seeks to address the equity gap. By opening research opportunities to all students as part of the regular curriculum for a degree, regardless of the individual circumstances of students, CUREs help level the playing field of high-impact practices. They give large numbers of students the opportunity to engage in research (Ahmad & Al-Thani, 2022; Gentile et al., 2017); something only a few students, and disproportionately not students from underrepresented minorities, otherwise benefit from through limited numbers of research internships (Bell et al., 2017; Desai et al., 2008; Gin et al., 2022; Haeger et al., 2015; Jayabalan et al., 2021; Jones & Lerner, 2019; Mahatmya et al., 2017; McLean et al., 2021; Shanahan, 2018; Stoeckman et al., 2019). By introducing a diverse body of students to research and its benefits, CUREs can spark interests in research careers, change students' identity, and contribute to changing the demographic and identity make-up of the research community over the longer term (Bangera & Brownell, 2014; Newell & Ulrich, 2022; Villarejo et al., 2008; Wilczek et al., 2022). Research experiences have been identified as one of the critical predictors of the persistence of underrepresented students in the sciences (Schultz et al., 2011; Villarejo et al., 2008). CUREs themselves have been showed to lead to increased engagement in research, and persistence in STEM (Bascom-Slack et al., 2012; Harrison et al., 2011; Harvey et al., 2014). The integration of research into the regular course load of students and the concentration of the work associated with the research project into the class time (or in replacement of regular homework) helps overcome many obstacles encountered by research internships in diversifying the research community including awareness of research opportunities, awareness of benefits of research experiences, unconscious societal bias, financial constraints, and personal barriers (Longmire-Avital 2018; Bangera & Brownell, 2014; Shanahan, 2018; Villarejo et al., 2008). In fact, CUREs have sometimes been showed to positively impact student groups traditionally underrepresented in the sciences specifically (Hanauer et al., 2022). CUREs have also been called the most complete intervention possible to address the hidden curriculum (Jayabalan et al., 2021). As such, they have the potential to participate in creating inclusive college classrooms, reduce the equity gap, and stem the leaks of underrepresented scholars (Handelsman et al., 2022).

CHAPTER OVERVIEW

2: WHY AND HOW? DEVELOPING YOUR OWN CURE

2.1: Why implement a CURE in your class 2.1: Why implement a CURE in your classEdit Update titlecancel

C) Why implement a CURE in your class

Jonathan J.-M. Calède

High-impact practices are associated with learning gains for students and benefits to the university community that include gains in GPA, increased student retention, improved student-instructor interactions, and more supportive campus environments (Kuh, 2008). Research experiences specifically have been demonstrated to benefit students' knowledge gains, skill acquisition, motivation, identity as researchers, comfort, and engagement (e.g., Follmer et al., 2017; Hanauer et al., 2017; Hunter et al., 2007; Landrum & Nelsen, 2002; Lopatto, 2007; Newell & Ulrich, 2022), including, and sometimes particularly, for students from underrepresented minorities (Daniels et al., 2016; Genet, 2021; Hanauer et al., 2017, 2022; Lopatto, 2007; Malotky et al., 2020; Martin et al., 2021; Matyas et al., 2022; Rodenbusch et al., 2016; Shuster et al., 2019), contributing to reducing the equity gap (Shapiro et al., 2015). CUREs are one tool that contributes to implementing a more equitable model of education moving forward (Elgin et al., 2021).

There is an extensive literature on the benefits of research experiences and CUREs, particularly for students. However, much of this literature is composed of studies that do not control for student-level characteristics, which could explain differences between groups (e.g., GPA, level of preparedness, enrollment preferences, prior research experience). In fact, many studies consist of pre-post comparisons of single groups and do not include comparison groups. Additionally, most of the assessments (e.g., quizzes, tests, and exams) of the content knowledge that was undertaken in published studies was subject-specific, preventing comparisons across learning experiences. Finally, analyses of skill gains and personal development overwhelmingly rely on self-reporting by students (Linn et al., 2015). In addition to cultural and identity issues associated with self-reporting and confidence, it can be difficult to determine whether students have indeed improved in a particular skill through the experience or merely became more confident about their skills (Dolan, 2016). Because CUREs are demonstrated to benefit self-efficacy (Martin et al., 2021), this type of data should be considered with care.

1. **Benefits to students and learning gains**

CUREs are widely recognized to benefit students. Most of the research is based on pre-post comparisons of self-reported levels of confidence, but numerous studies also include knowledge assessments using quizzes, comparisons with traditional laboratory or recitations sections employing "cookbook" labs and activities, and comparisons with mentored research experiences. There are some analyses that fail to identify strong signals of improved skills (e.g., Brownell et al., 2015), but for many aspects of the research experience, student gains in CUREs meet or exceed those observed in summer internship and mentored student research experience models (Bixby & Miliauskas, 2022; Evans et al., 2021; Frantz et al., 2006; Hanauer et al., 2012; Jordan et al., 2014; Lopatto et al., 2008; Overath et al., 2016; Shaffer et al., 2010; Shapiro et al., 2015; Smith et al., 2021). Student gains in CUREs are higher than those experienced by students in traditional labs (Blumling et al., 2022; Brownell et al., 2012; Evans et al., 2021; Hanauer et al., 2017; Jordan et al., 2014; Lopatto et al., 2008; Newell & Ulrich, 2022; Pavlova et al., 2021; Pontrello, 2015; Wolkow et al., 2014; Wu & Wu, 2022). Analyses of CUREs across institutions, disciplines, and course levels show that the greatest gains are found in CUREs that are implemented over the course of an entire semester, allow for student input in the research process, and focus on novel research with outcomes unknown to both instructors and students (DeChenne-Peters et al. 2023; Mader et al., 2017). Additionally, students enrolling in a course-sequence including more than one CURE see additional gains (Corwin et al., 2022). There have also been reports that student gains are higher or more critical in CUREs at the introductory level than the upper level (Hanauer et al., 2017; Handelsman et al., 2022; Ruttledge, 1998).

Broadly speaking, the impacts of CUREs on students can be assigned to five categories:

1. Gains in content knowledge and technical skills
2. Gains in broadly applicable skills
3. Changes in attitudes towards and understanding of research
4. Gains in confidence and self-efficacy
5. Changes in professional/career paths

Gains in content knowledge and technical skills

One of the strongest gains experienced by students in CUREs is an improved content knowledge. This impact is observed across disciplines including CUREs in molecular biology (Harvey et al., 2014; Makarevitch et al., 2015), plant biology (Ward et al., 2014), ecology (Valliere, 2022a, 2022b), microbiology (DeHaven et al., 2022), geosciences (Gonzales & Semken, 2009; Kortz & van der Hoeven Kraft, 2016), genomics (Drew & Triplett, 2008), ecology (Genet, 2021), and cell biology as well as genetics (Makarevitch et al., 2015; Siritunga et al., 2011). This increased content knowledge can translate to improved grades through time or compared to non-CURE course sections (Blumling et al., 2022; Ing et al., 2021; Jordan et al., 2014; Olimpo et al., 2016; Shaffer et al., 2010; Waynant et al., 2022; Winkelmann et al., 2015). It may also participate in increasing student retention within courses (Blumling et al., 2022) as well as between first and second years (Jordan et al., 2014). Importantly, the time-constraints of CUREs on content coverage does not have a detrimental effect on students' understanding of biological concepts beyond the focus of the CURE (Jordan et al., 2014) or overall content knowledge in the discipline (Wolkow et al., 2014). Several assessments of CUREs across disciplines have also found strong gains in students' technical skills (Kortz & van der Hoeven Kraft, 2016) including in computer modelling, software program use, and general computer use (Drew & Triplett, 2008; Pavlova et al., 2021; Williams & Reddish, 2018), enhanced statistical knowledge (Olimpo et al., 2018; Pavlova et al., 2021; Ward et al., 2014), and laboratory techniques (Bixby & Miliauskas, 2022; Evans et al., 2021; Jordan et al., 2014; Large et al., 2022; Siritunga et al., 2011; Stoeckman et al., 2019). CUREs also improve students' ability and confidence in their ability to design experiments and interpret data (Bixby & Miliauskas, 2022; Blumling et al., 2022; Brownell et al., 2012; Genet, 2021; Kloser et al., 2013; Large et al., 2022; Martin et al., 2021; Pavlova et al., 2021; Shaffer et al., 2014).

Gains in broadly applicable skills

Students also widely report great gains in soft skills. Those skills are applicable to courses and experiences through the student's academic career and beyond in their professional careers; they are not limited to the discipline associated with the CURE. Thus, students indicate a greater familiarity with the structure of scholarly papers and a greater ability to engage with the primary literature following engagement in a CURE (e.g., DeHaven et al., 2022; Drew & Triplett, 2008; Evans et al., 2021; Jordan et al., 2014; Martin et al., 2021; Shelby, 2019; Valliere, 2022b). Several studies also report improved written and oral communication, including the graphical representation of data and the presentation of both the research process and research findings (DeHaven et al., 2022; Genet, 2021; Jordan et al., 2014; Large et al., 2022; Makarevitch et al., 2015; Shaffer et al., 2014; Shelby, 2019; Stoeckman et al., 2019; Valliere, 2022a, 2022b; Ward et al., 2014; Wiley & Stover, 2014; Williams & Reddish, 2018). This improved ability to communicate is associated with increased willingness and confidence in communicating research (e.g., Kloser et al., 2013; Valliere, 2022a). There is also evidence that CUREs lead to improved time management and organization skills (Kortz & van der Hoeven Kraft, 2016) as well as increased problem-solving skills (Olimpo et al., 2016; Wu & Wu, 2022). Students engaged in CUREs gain an appreciation for the obstacles and challenges of research (Drew & Triplett, 2008). In fact, a major impact of CUREs is their ability to increase tolerance for obstacles in students (Corwin et al., 2022; Evans et al., 2021; Jordan et al., 2014; Large et al., 2022; Stoeckman et al., 2019; Williams & Reddish, 2018; Wu & Wu, 2022). Students report valuing the ability to learn from their mistakes in the CURE (Harrison et al., 2011). Even when research goals are not met, students engaged in CUREs increase their ability to navigate research obstacles (Gin et al., 2018). CUREs lead students to become more active learners who can better think independently, are motivated to learn, and better able to think in new ways (Evans et al., 2021; Harrison et al., 2011; Kortz & van der Hoeven Kraft, 2016; Shaffer et al., 2010). Research has also showed that students value the potential for publication of the research they engage in as part of the CURE (Wiley & Stover, 2014); these publications may facilitate future admission into graduate and professional programs.

Changes in attitudes towards and understanding of research

CUREs are an enjoyable experience for students (Carr et al., 2018; Drew & Triplett, 2008; Harvey et al., 2014; LaForge & Martin, 2022; Pontrello, 2015). Students like the experience of exploring an open-ended question with no known outcome (Brownell et al., 2012; Hanauer et al., 2012; Harrison et al., 2011; Williams & Reddish, 2018), making decisions in their work (Hanauer et al., 2012; Harrison et al., 2011), and the relevance of their work to the scholarly community and the world (Drew & Triplett, 2008; Hanauer et al., 2012; Jordan et al., 2014; LaForge & Martin, 2022; Tomasik et al., 2013). Students often find CUREs to be community-building (e.g., Kulesza et al., 2022; Large et al., 2022; Hanauer et al., 2017; Werth et al., 2022) and recognize the positive impact of CUREs on their professional paths (e.g., Amir et al., 2022). CUREs also contribute to a better understanding of and confidence in the research process and science (Bascom-Slack et al., 2012; Bixby & Miliauskas, 2022; Evans et al., 2021; Freeman et al., 2023; Hanauer et al., 2012; Harrison et al., 2011; Jordan et al., 2014; Kulesza et al., 2022; LaForge & Martin, 2022; Large et al., 2022; Shaffer et al., 2014; Stoeckman et al., 2019). Harrison et al. (2011) reported increased interests in science and research in students who took a CURE. Valliere (2022a) documented a significant increase in students' perception that they can personally relate to a scientist. Kortz and van der Hoeven Kraft (2016) found an increased general appreciation for science and scientists in students engaged in CUREs. This positive attitude towards research is retained for years (Harvey et al., 2014). CUREs also lead students to identify as researchers or as members of the scholarly community and not merely students more than they do after traditional courses (Hanauer et al., 2017; Mraz-Craig et al., 2018). Additionally, students involved in a CURE were found to have developed a better understanding of the distinctions between hypotheses and theories as well as a deeper grasp of the importance of creativity in research compared to students who completed a traditional lab course (Russell & Weaver, 2011). Interestingly, Dewey et al. (2022) found that different CURE models lead students to perceive scientific research in different ways and identify different elements as central to the culture of research in the discipline. Designing an inclusive CURE experience is therefore critical to the experience of the students and their view of research.

Gains in confidence and self-efficacy

Because of the nature of the data collection undertaken in evaluating many CUREs, much of the information collected on the benefits of CUREs concerns the confidence of students with specific tasks and general self-efficacy. Kortz and van der Hoeven Kraft (2016) reported increased student confidence in talking with other people and more open minds in students. In fact, an increased willingness and ability to engage in conversations and collaborations is a common outcome of CUREs (Brownell et al., 2012; Jordan et al., 2014; Martin et al., 2021; Stoeckman et al., 2019; Vater et al., 2021). Yet, CUREs also promote the ability to work independently (Stoeckman et al., 2019; Jordan et al., 2014). More broadly, students report greater confidence with their ability to conduct research (Fendos et al., 2022; Jordan et al., 2014; Siritunga et al., 2011, Wu & Wu, 2022) and self-efficacy in general (e.g., Hanauer et al., 2017).

Changes in professional/career paths

CUREs have a significant effect on students' identity as scholars, particularly in the sciences, boosting their intention to pursue a career in STEM (Newell & Ulrich, 2022). In fact, for many students, CUREs help clarify their career path (Jordan et al., 2014; Shaffer et al., 2014; Stoeckman et al., 2019). CUREs increase the matriculation in STEM majors (Rodenbusch et al., 2016). They also directly or indirectly lead to increased graduation rates (Rodenbusch et al., 2016; Waynant et al., 2022). Additionally, several studies have found that engagement in a CURE leads to increased interest in conducting research in other settings following the course (Bascom-Slack et al., 2012; Brownell et al., 2012; Carr et al., 2018; Fendos et al., 2022; Harvey et al., 2014; Overath et al., 2016; Shaffer et al., 2014; Ward et al., 2014). Students also mention feeling better prepared to undertake research projects following their engagement in a CURE (Bascom-Slack et al., 2012; Drew & Triplett, 2008; Jordan et al., 2014; Newell & Ulrich, 2022; Stoeckman et al., 2019; Williams & Reddish, 2018). This leads to greater engagement in traditional research experiences (Harvey et al., 2014). This interest in research also translates to changes in career paths with increased interests and matriculation in graduate school and medical school (Harrison et al., 2011; Bascom-Slack et al., 2012; Shaffer et al., 2014). Many years later, CUREs lead to increased retention in scientific careers (Harvey et al., 2014; Shaffer et al., 2014).

2. **Benefits to instructors**

CUREs provide instructors with opportunities to improve student learning, help undergraduates develop a portfolio of works that help them reach their career goals, and foster chances for students to gain skills (Desai et al., 2008; Lopatto et al., 2014; see also section C1 above). As such, CUREs enable instructors, postdoctoral researchers, and graduate teaching assistants to pursue the mission of their institution as well as their own educational goals.

CURE instructors may be faculty members, instructors, postdoctoral researchers, or graduate student assistants (Goodwin et al., 2021; Heim & Holt, 2019). Cascella & Jez (2018) presented the argument that CUREs represent a great opportunity to train postdoctoral researchers and graduate students in the roles of instructors and principal investigator. They specifically argue that CUREs provide the opportunity to develop instructional materials, practice active learning methods of teaching, and build an identity and practice as a teacher beyond merely assisting in grading and delivering content (Cascella & Jez, 2018). The nature of CUREs also provides a platform for trainees to become familiar with the functioning and management of large research projects that involve personnel, deadlines, and a budget (see also Desai et al., 2008 for a similar argument in a learning community context). Because of the potential for publication of the results of the original research undertaken in CUREs, trainees also maintain or increase their productivity (in the form of presentations, publications, or the generation of data that can feed into proposals or manuscripts). Little research has been undertaken assessing the experience of graduate teaching assistants (GTAs) in CUREs. Heim and Holt (2019) provided limited data that support the conclusion that GTAs value the experience of mentoring undergraduate research, but many report the experience being very challenging. Goodwin et al. (2021) similarly found that graduate student instructors almost universally recognize the pedagogical value and benefits of CUREs (for both students and GTA), but that many report high costs to teaching CUREs, including time and emotional investment. In fact, in an important link between benefits to undergraduates and benefits to instructors in training, research shows that the individual experiences of students across course sections taught by different GTAs vary widely (Goodwin et al., 2022, 2023). These experiences appear to be affected by the beliefs, motivation, interests, training, and attitude of the GTAs more than their research and teaching experiences (Goodwin et al., 2022, 2023). Specifically, the ability of GTAs to provide support necessary for undergraduate students to persevere through failures and repeat their work is likely an important predictor of the quality of the students' research experience (Goodwin et al., 2022). The motivation of undergraduate students taking the CURE, in particular, appears to be highly affected by GTA training and practice (Goodwin et al., 2023). Support from instructors of record, peers, and undergraduate teaching assistants is therefore critical to the success of GTAs teaching a CURE as well as the success of their students (Goodwin et al., 2021, 2022, 2023). Kern and Olimpo (2022) have developed an effective training program to help GTA facilitate CUREs.

There are also many benefits of teaching a CURE for principal investigators and lecturers. A survey of 16 instructors across disciplines at the Ohio State University who have implemented a CURE shows a range of positive impacts. The open answers of the instructors were categorized in each of eight categories. Instructors were able to identify more than one benefit of teaching a CURE resulting in 24 coded impacts (**Table 2**). Three instructors did not identify a benefit (responding "unclear" or "not sure").

Benefit	N
Increased research productivity	4
Growth of research group	2
Personal satisfaction and fulfillement	5
Increased university community engagement	2
Increased student engagement and enjoyment	5
Increased recruitment/retention of students in major/program	2
Personal learning	1
Pedagogical development	3

Table 2. Benefits of implementing a CURE identified by 13 instructors at the Ohio State University.
N is the frequency of instructors for each category.

Below, I focus on the benefits of CUREs to instructors that positively impact the career of instructors and are best characterized as self-interests (Desai et al., 2008). Although much less research has been undertaken on instructor gains than students', CUREs have widely been recognized to benefit instructors. Instructor impacts can be divided into three categories:

1. Engaging in a meaningful research-driven teaching practice
2. Boosting engagement in research and increasing productivity
3. Gaining access to resources and developing a network of colleagues

Engaging in a meaningful research-driven teaching practice

CUREs provide the opportunity for instructors to integrate their research and teaching missions (Fukami, 2013; Shortlidge & Brownell, 2016). This includes the pursuit of one's research program as well as the chance to go into new directions and satisfy one's intellectual curiosity (Desai et al., 2008; Roberts and Shell, 2023; Shortlidge et al., 2016). As such, CUREs are more enjoyable to teach than traditional labs (Shortlidge et al., 2016; DeChenne-Peters & Scheuermann, 2022) and enable instructors to meaningfully engage students in the discipline without the barrier of content-coverage (Elgin et al., 2021). Instructors report that CUREs enable them to improve their pedagogical knowledge and develop an interest in the formal assessment of their own teaching (Craig, 2017). CUREs also enable instructors to teach in a way that promotes student enthusiasm and motivation (Lopatto et al., 2014). Some instructors report that CUREs improve their relationships with students (Shortlidge et al., 2016). Others report that CUREs improve their job satisfaction (Shortlidge et al., 2017). CUREs can be components of broader impacts of grant proposals and thus support the success of both research and teaching in one more way (see Shortlidge et al., 2016).

Boosting engagement in research and increasing productivity

For instructors at primarily undergraduate institutions and lecturers at research universities whose faculty model involves high teaching loads, CUREs offer the opportunity to remain involved in research endeavors (Hewlett, 2018; DeChenne-Peters & Scheuermann, 2022) and provide research opportunities to students when uncommitted time available for mentored research experiences is lacking (Gentile et al., 2017). Instructors benefit from the opportunity to keep up with the field of research and the literature (Lopatto et al., 2014). CUREs also increase the confidence of some faculty members in their own research (Shaffer et al., 2010). Because CUREs enable instructors to explore significant research questions that are broadly relevant to the scholarly community, they can be beneficial in generating data for faculty research (Shortlidge et al., 2016). In fact, the significance of the research itself has been found to be a motivator for many instructors (Lopatto et al., 2014), just like for students. Numerous faculty members have reported productivity benefits from CUREs, particularly co-authorship on publications (Lopatto et al., 2014, Shortlidge et al., 2016). Publication is an important outcome and motivator of engagement in the development and implementation of CUREs for both faculty members and students (e.g., Hatfull et al., 2006; Jordan et al., 2014; Ward et al., 2014). For some researchers, CUREs also offer the opportunity to identify and recruit trained and motivated students for mentored research internships (Overath, 2016; Shortlidge et al., 2016; Stoeckman et al., 2019; Ott et al., 2020; Elgin et al., 2021), or even prepare students for mentored research experiences (Fendos et al. 2022). The many research and teaching gains of CUREs lead some instructors to report increased prestige or improved reputation (Lopatto et al., 2014; Shaffer et al., 2010). CUREs are influential for promotion and tenure (Shortlidge et al., 2016).

Gaining access to resources and developing a network of colleagues

Surveys of instructors who have implemented CUREs show their impacts go beyond direct career benefits. CURE instructors value the chance to gain access to new technology on campus and beyond (Shaffer et al., 2010). Involvements in CUREs, particularly when instructors join national collaborative efforts like the Genomics Education Partnership (Lopatto et al., 2008), the Biological Collections in Ecology and Evolution Network (https://bceenetwork.org/), the Malate Dehydrogenase CURE Community (Provost 2022), or the Science Education Alliance Phage Hunters Advancing Genomics and Evolutionary Science (SEA-PHAGES; Jordan et al., 2014) also enable instructors to gain colleagues, grow their network of collaborators, and connect to a larger research and teaching community (Lopatto et al., 2014; Shaffer et al., 2010; DeChenne-Peters & Scheuermann, 2022). Instructors involved in CUREs report professional growth from the experience (Lopatto et al., 2014).

3. **Benefits to the institution**

Many of the benefits of CUREs to the institution are encapsulated in the benefits to instructors and students. Institutions benefit from higher student retention and graduation rates, training a more diverse workforce, and developing a more inclusive learning environment. They also gain from having happier instructors with higher research outputs and potentially higher tenure rates, who are less likely to leave their positions (Elgin et al., 2021; Shortlidge et al., 2017). CUREs can in fact contribute to increased enrollment (Bell et al., 2017). Similarly to service-learning courses, CUREs can provide institutions the opportunity to partner with business and community organizations (Elgin et al., 2021; Malotky et al., 2020; Silvestri, 2018) and provide actionable information for the students' community (Smith et al., 2022; Valliere, 2022a) or partners around the world (Kay et al., 2023). CUREs have also been showed to inspire faculty members to seek grant support for research and education (Shaffer et al., 2010), which can bring the institution additional funds through indirect-cost recovery. The CUREs themselves or the outputs from these CUREs can also participate in increasing the prestige of the institution with an increased number of publications, presentations, and awards (e.g., Ahmad & Al-Thani, 2022; Shaffer et al., 2010; Overath, 2016; Bell et al., 2017). Recent calls for institutions to support the development and implementation of CUREs have emphasized their importance in overcoming the opportunity gap in college (e.g., Handelsman et al., 2022).

2.2: Designing a successful CURE- The basics 2.2: Designing a successful CURE- The basicsEdit Update titlecancel

D) Designing a successful CURE: The basics

Jonathan J.-M. Calède

Although CUREs share several characteristics (see Introduction), there exists a diversity of CURE designs and models. Shuster et al. (2019) proposed a taxonomy of CUREs with two main categories: researcher-independent CUREs and researcher-driven CUREs. The former category includes many discovery-based CUREs. Researcher-independent CUREs are not tied to the expertise or research interests of the instructor. As such, they can be supervised by different instructors, may have a greater lifespan, and can potentially foster student-driven questions (Shuster et al., 2019). However, these CUREs are supervised by non-expert researchers who may not be confident in the project and require training (Shuster et al., 2019). Researcher-independent discovery-based CUREs can be replicated across institutions and lead to national programs (e.g., Genné-Bacon & Bascom-Slack, 2018; Jordan et al., 2014; Shaffer et al., 2014). Some of those national programs have common research goals. For example, the Genomics Education Partnership (GEP) is annotating the genome of *Drosophila* (Shaffer et al., 2010; 2014). Instructors can apply to join these programs and receive centralized training, benefit from the support system put in place by these project's networks, and exchange with the associated communities of instructors and researchers. One advantage of such programs is the potential time savings made by instructors in developing and initiating a CURE (Gentile et al., 2017). There are also national programs that have been developed to support students and their instructors in undertaking their own research projects. Examples of these programs include The Genome Consortium for Active Teaching (GCAT; Campbell et al., 2007; Walker et al., 2008) and its sister program, GCAT-SEEK (Buonaccorsi et al., 2014; Buonaccorsi et al., 2011) as well as the Ecological Research as Education Network. Both GEP and GCAT have been extensively analyzed for their ability to support CUREs (Lopatto et al., 2014). Malotky et al. (2020) present yet another model of researcher-independent CURE in their CEL-CURE. This CURE involves "Community-Engaged Learning"; the research questions investigated by the students each semester stem from conversations with community partners in an original combination of service-learning and research (Malotky et al., 2020). Similarly, Adkins-Jablonsky et al. (2020) developed a series of CUREs centered around environmental justice in an effort to increase community-engagement as well as science efficacy and identity.

Researcher-driven CUREs are experiences in which the students contribute to the research program of the instructor. Researcher-driven CUREs are mentored by experts who are invested in the success of the CURE and its dissemination to the research community (Shuster et al., 2019). These CUREs are likely to be hypothesis-driven and lead to publication (Fukami, 2013; Shuster et al., 2019). There are some opportunities for researcher-driven CUREs to benefit from the support of national programs–the Keck Geology Consortium provides such an opportunity (De Wet et al., 2009)–but these opportunities are rare and often support small numbers of students (six to nine students in the case of the Keck Consortium). Because of the inherent integration of their teaching and research missions in researcher-driven CUREs, faculty members at research institutions may be more likely to be invested in the development and success of such CUREs as opposed to researcher-independent ones, whereas faculty members at community colleges and institutions with little support for the course development efforts associated with creating a CURE, and limited research funding, can benefit from joining national programs that provide a framework for implementing CUREs.

Thus, there are three possible paths to designing a CURE:

1. Implementing a researcher-independent CURE for which there is a pre-existing structure available through national programs or the peer-reviewed discipline-based education research (DBER) literature
2. Developing a new researcher-independent CURE
3. Designing a unique researcher-driven CURE

When designing a new CURE, just like any other course, instructors should adopt a backward-design approach (Dolan, 2016; Graff, 2011; Hills et al., 2020; Shapiro et al., 2015; Wiggins & McTighe, 1998). This approach requires instructors to first identify the outcomes desired from the CURE, then determine the acceptable evidence that these outcomes have been met, and only afterwards, plan the learning experiences and instruction (Cooper et al., 2017). One important caveat to this structure when designing a researcher-driven CURE is the need to integrate the research goals of the researcher with the learning outcomes for the students. Such integration can and should follow a backward-design approach for the research component as well (Cooper et al., 2017: Table 1). It may be necessary to adjust the research project to meet the desired learning outcomes or revise the learning outcomes of the CURE to match the limitations of the research project (Cooper et al., 2017). The instruction and mentorship of the research experience itself will need to consider several critical issues (e.g., Cooper & Brownell, 2018; Kloser et al., 2011; Shaffer et al., 2014; Zelaya et al., 2020) presented in **Table 3**.

Logistical issues	Duration of the experience
	Equipment and supply needs including IT resources, lab equipment, and reagents
	Class size and space availability
	Staff and teaching assistant support
Research issues	Technical expertise of the students and the data collection/analytical needs of the research project
	Expertise of the instructor with the study system
	Availability and logistics (including permitting) of specimens, resources, or organisms for the data collection
	Size and complexity of the dataset, the number of variables, as well as the diversity of hypotheses and questions the students will be able to explore
	Setup of checks and balances for student-collected data
Pedagogical issues	Integration of program requirements
	Scaffolding of the process of research
	Feedback structure including instructor-feedback and peer-reviews
	Format of the final deliverable and presentation of the findings
	Accessibility and inclusivity

Table 3. Critical issues to consider in developing a CURE.

The issues of **Table 3** can be rephrased as framing questions (questions 1-2, 4-5, 7-8, and 10 from Dolan, 2016) to guide the development of a CURE:

1. *How will the CURE be integrated into the curriculum?* Identifying the target audience of the course, the nature of the course, and its place within the curriculum as well as the CURE's research goals.
2. *To what extent will students have intellectual responsibility and [ownership] of the research?* Determining the role the students will have in developing, implementing, and communicating the research.
3. *Which components of the research process will be integrated into the CURE?* Defining the research experience of the students and guiding them through the literature, data collection, and data analysis.
4. *How will research progress be balanced with student learning and development?* Designing an inclusive experience that includes peer-reviews and addresses the limitations imposed by the course structure.
5. *How will the research learning tasks be structured to foster students' development as [scholars]?* Scaffolding the CURE and individual assignments, fostering reflection, and promoting successful group work.
6. *How will students communicate the results of their research?* Choosing the appropriate mode of communication to reflect authentic scholarship and ensure equity.
7. *How will [the progress and experience of students] be assessed?* Adopting a mode of grading that is true to the research process as well as transparent and inclusive for all students.
8. *How will research learning tasks change as discoveries are made and initial research questions are answered?* Including iteration within the course and ensuring the success of a CURE over the long term.
9. *What are the logistical obstacles and solutions for the different steps of the CURE?* Overcoming problems accessing and analyzing data and funding a CURE.
10. *What are the roles of instructional [and support] staff?* Transforming instructors into mentors to support students and teaching assistants.
11. *How will the success of the CURE be assessed?* Assessing the usefulness of the scaffold of the CURE and its individual activities and evaluating the success of the research and the students.

1. **Successful CUREs across disciplines**

Many successful CUREs have been developed across institutions, course-levels, and disciplines. **Table 4** presents a selection of CUREs, sorted by discipline, focusing on examples that provide templates or curriculum elements for the replication of the experience or its implementation in a different context. These examples can be used to start reflecting on the questions presented above.

Examples of CUREs across disciplines and topic

References for CUREs at the introductory and/or upper level in Anthropology, Biology, Business, Chemistry, Criminal Justice, Engineering, Forensic Science, Geosciences, Human Resource Development, Information security, Linguistics, Mathematics, Psychology, Physics, as well as Writing and Composition.

Discipline	Level	Topic	Reference
Anthropology	Both	Equity, Health, Obesity, etc.	Ruth et al., 2021
Biology	Both	Botany	Ward et al., 2014
Biology	Both	Conservation Biology	Sorensen et al., 2018
Biology	Both	DBER	Cooper & Brownell, 2018
Biology	Both	DBER	Mohammed et al., 2021

Discipline	Level	Topic	Reference
Biology	Both	Ecology	Russell et al., 2015
Biology	Both	Genomics – GEP	Lopatto et al., 2008
Biology	Both	Genomics – GEP	Shaffer et al., 2010
Biology	Both	Genomics – GEP	Shaffer et al., 2014
Biology	Both	Microbiology	Adkins-Jablonsky et al., 2020
Biology	Both	Microbiology	Lyles & Oli, 2022
Biology	Both	Microbiology	Zelaya et al., 2020
Biology	Both	Molecular Biology	Russell et al., 2015
Biology	Both	Molecular Biology/Ecology…	Poole et al., 2022
Biology	Both	Public Health	Malotky et al., 2020
Biology	Intro	Developmental Biology	Sarmah et al., 2016
Biology	Intro	Ecological Genetics	Bucklin & Mauger, 2022
Biology	Introductory	Botany	Murren et al., 2019
Biology	Introductory	Ecology	Brownell et al., 2012
Biology	Introductory	Ecology	Fukami, 2013
Biology	Introductory	Ecology	Genet, 2021
Biology	Introductory	Ecology	Kloser et al., 2011
Biology	Introductory	Ecology	Kloser et al., 2013
Biology	Introductory	Ecology	Young et al., 2021
Biology	Introductory	Field Ecology	Stanfield et al., 2022
Biology	Introductory	Field Ecology	Thompson et al., 2016
Biology	Introductory	Genetics	Brownell et al., 2015
Biology	Introductory	Genetics	Mills et al., 2021
Biology	Introductory	Genetics/Ecology/Evolution…	Bakshi et al., 2016
Biology	Introductory	Genomics	Bowling et al., 2016
Biology	Introductory	Genomics	Burnette & Wessler, 2013
Biology	Introductory	Genomics	Chen et al., 2005
Biology	Introductory	Genomics	Evans et al., 2021
Biology	Introductory	Genomics	Hatfull et al., 2006
Biology	Introductory	Genomics	Makarevitch et al., 2015
Biology	Introductory	Genomics	Wiley & Stover, 2014
Biology	Introductory	Genomics	Wolkow et al., 2014
Biology	Introductory	Genomics – SEA- PHAGES	Harrison et al., 2011
Biology	Introductory	Microbiology	Peyton & Skorupa, 2021
Biology	Introductory	Molecular Biology	Hekmat-Scafe et al., 2017
Biology	Introductory	Neuroscience	Waddell et al., 2021
Biology	Upper	Cell Biology	Shapiro et al., 2015
Biology	Upper	Cell Biology	Siritunga et al., 2011
Biology	Upper	Conservation Biology	Gastreich, 2020
Biology	Upper	Ecology	Shapiro et al., 2015
Biology	Upper	Genetics	Delventhal & Steinhauer, 2020
Biology	Upper	Genetics	Li et al., 2016
Biology	Upper	Genetics	McDonough et al., 2017
Biology	Upper	Genomics	Drew & Triplett, 2008

Discipline	Level	Topic	Reference
Biology	Upper	Genomics	Dunne et al., 2014
Biology	Upper	Genomics	Harvey et al., 2014
Biology	Upper	Genomics	Martin et al., 2020
Biology	Upper	Immunology	Cooper et al., 2019
Biology	Upper	Metagenomics	Baker et al., 2021
Biology	Upper	Microbiology	DeHaven et al., 2022
Biology	Upper	Microbiology	Jurgensen et al., 2021
Biology	Upper	Microbiology	Pedwell et al., 2018
Biology	Upper	Microbiology	Petrie, 2020
Biology	Upper	Microbiology	Sewall et al., 2020
Biology	Upper	Microbiology	Shapiro et al., 2015
Biology	Upper	Microbiology	Zelaya et al., 2020
Biology	Upper	Molecular Biology	Shanle et al., 2016
Biology	Upper	Molecular Biology	Shapiro et al., 2015
Biology	Upper	Molecular Biology	Shuster et al., 2019
Biology	Upper	Molecular Biology	Siritunga et al., 2011
Biology	Upper	Physiological Ecology	Ramírez-Lugo et al., 2021
Biology	Upper	Physiology	Rennhack et al., 2020
Biology	Upper	Restoration Ecology	Valliere et al., 2022b
Biology	Upper	Urban Ecology	Valliere et al., 2022a
Biology	Upper	Virology	Shapiro et al., 2015
Biology*	Introductory	Genomics – SEA- PHAGES	Jordan et al., 2014
Biology*	Introductory	Microbiology	Genné-Bacon & Bascom-Slack, 2018
Biology*	Introductory	Plant Microbiome	Bascom-Slack et al., 2012
Biology†	Introductory	Molecular Biology	Boltax et al., 2015
Biology†	Introductory	Molecular Biology	Rowland et al., 2012
Biology†	Introductory	Organismal Biomechanics	Full et al., 2015
Business	Upper	Retail	Sternquist et al., 2018
Chemistry	Both	Biochemistry	Roberts et al., 2019
Chemistry	Both	Biochemistry	Shelby, 2019
Chemistry	Both	Biochemistry	Vater et al., 2021
Chemistry	Introductory	Analytical Chemistry	Silvestri, 2018
Chemistry	Introductory	Biochemistry	Bell, 2011
Chemistry	Introductory	Biochemistry	Chaari et al., 2020
Chemistry	Introductory	Biochemistry	Knutson et al., 2010
Chemistry	Introductory	Bioremediation	Silsby et al., 2022
Chemistry	Introductory	General Chemistry	Blumling et al., 2022
Chemistry	Introductory	General Chemistry	Miller et al., 2022
Chemistry	Introductory	General Chemistry	Tomasik et al., 2013
Chemistry	Introductory	General Chemistry	Weaver et al., 2006
Chemistry	Introductory	General Chemistry	Winkelmann et al., 2015
Chemistry	Introductory	Organic Chemistry	Alaimo et al., 2014
Chemistry	Introductory	Organic Chemistry	Carr et al., 2018
Chemistry	Introductory	Organic Chemistry	Cruz et al., 2020

Discipline	Level	Topic	Reference
Chemistry	Introductory	Organic Chemistry	Pontrello, 2015
Chemistry	Introductory	Organic Chemistry	Ruttledge, 1998
Chemistry	Introductory	Organic Chemistry	Silverberg et al., 2018
Chemistry	Introductory	Organic Chemistry	Weaver et al., 2006
Chemistry	Introductory	Organic Chemistry	Wilczek et al., 2022
Chemistry	Upper	Analytical Chemistry	Tomasik et al., 2014
Chemistry	Upper	Biochemistry	Ayella & Beck, 2018
Chemistry	Upper	Biochemistry	Colabroy, 2011
Chemistry	Upper	Biochemistry	Large et al., 2022
Chemistry	Upper	Biochemistry	Satusky et al., 2022
Chemistry	Upper	Biophysical Chemistry	Hati & Bhattacharyya, 2018
Chemistry	Upper	Medicinal Chemistry	Williams & Reddish, 2018
Chemistry	Upper	Polymer Chemistry	Karlsson et al., 2022
Chemistry*	Introductory	General Chemistry	Clark et al., 2016
Chemistry†	Introductory	Biochemistry	Rowland et al., 2012
Chemistry†	Introductory	Organic Chemistry	Boltax et al., 2015
Criminal Justice	Introductory	Crime statistics	Kruis et al., 2022
Criminal Justice	Introductory	Crime statistics	McLean et al., 2021
Engineering	Introductory	Biofluid Mechanics	Clyne et al., 2019
Engineering	Upper	Software Development	Abler et al., 2011
Engineering†	Introductory	Organismal Biomechanics	Full et al., 2015
Forensic Science	Upper	Next Generation Sequencing	Elkins & Zeller, 2020
Geosciences	Introductory	DBER	Kortz & van der Hoeven Kraft, 2016
Geosciences	Upper	Field Geomorphology	May et al., 2009
Geosciences	Upper	Field Glaciology	Connor, 2009
Geosciences	Upper	Field Petrology	Gonzales & Semken, 2009
Human Resource Development	Upper	Any	Hwang & Franklin, 2022
Information Security	Upper	Vulnerability Scanning	Xu et al., 2022
Linguistics	Introductory	Phonetics and Phonology	Bjorndahl & Gibson, 2022
Mathematics	Both	Any	Deka et al., 2022
Physics	Introductory	Solar Physics	Werth et al., 2022
Psychology	Upper	Cognitive Neuroscience	Wilson, 2022
Psychology	Upper	Music Psychology	Hernandez-Ruiz & Dvorak, 2020
Writing & Composition	Introductory	Oral History	Parsons et al., 2021
Writing & Composition	Introductory	Writing Pedagogy	Kao et al., 2020

Table 4. Examples of CUREs across disciplines and topic. * CURE implemented at the Ohio State University, † Interdisciplinary CURE

In addition to the many published CUREs, there are also several CUREs that have been developed across the campuses of Ohio State in STEM, social sciences, and humanities. **Table 5** provides examples of these courses at the time of writing including the instructor or contact person who may be able to provide documents from their course and/or insights into their experience developing and implementing the CURE.

Examples of unpublished CUREs across disciplines and topic at the Ohio State University

Information for select CUREs in STEM, Social sciences, and Humanities at the Ohio State University

Discipline	Course	Topic	Contact Information
Microbiology	Micro 2100	Yeasts and Fermentation	Steven Carlson: carlson.271@osu.edu
Evolutionary Biology	EEOB 4220	Mammal Ecology and Evolution	Ryan Norris: norris.667@osu.edu

Discipline	Course	Topic	Contact Information
Comparative Studies	CS 5189-S	Field Ethnography	Katherine Borland: borland.19@osu.edu, Jasper Waugh-Quasebarth: waugh-quasebarth.1@osu.edu
Women’s, Gender, and Sexuality Studies	WGSST 2550	History of Feminist Thought	Mytheli Sreenivas, sreenivas.2@osu.edu
History	HIST 2475	History of the Holocaust	Robin Judd, judd.18@osu.edu
English	ENG 4523	Renaissance London: Literature, Culture, and Place, 1540-1660	Chris Highley, highly.1@osu.edu

Table 5. Examples of unpublished CUREs across disciplines and topic at the Ohio State University.

2. **Expected learning outcomes of CUREs**

CUREs can help foster student success in different components of the curriculum. They can be implemented at the introductory level as well as in upper classes (**Table 3**) and even in first-year seminars (Vater et al., 2019); they can lead to thesis projects and fulfill other writing requirements; they can also involve extensive laboratory work (e.g., Pontrello, 2015), field work (e.g., Gonzales & Semken, 2009; Messager et al., 2022; Thompson et al., 2016), or community-based interventions (e.g., Malotky et al., 2020). As such, CUREs may be appropriate as entire classes or elements of classes in foundation courses, theme courses, honors classes, non-major courses, required introductory courses for the major, or upper-level electives.

The first step in the design of a CURE is to identify the desired pedagogical goals for the course. Keep in mind that pedagogical goals, or learning goals, describe what you as instructor, and your program, aim for with the CURE. They give students a general idea of what they will gain from the CURE. Expected learning outcomes (ELOs) describe what students are able to do at the outset of the CURE. ELOs are specific statements that use action verbs to state what student should achieve; they should be measurable and realistic.

There are three important categories of learning goals to consider. The first one concerns discipline-specific knowledge and skills, including technical skills. The second one is concerned with soft skills, including broadly applicable competences in communication and habits that promote success, like the ability to work well as part of a team. The third category of learning goals that is worth including in the framework of a CURE is personal goals. Encouraging students to develop their own learning goals for the CURE can be a powerful way to increase buy-in and ownership of the project (see [**Activity 1]**). It has also been linked to more positive impacts of the course for students (Lopatto et al., 2022). Whichever of these three categories they belong to, the goals should be aligned with appropriate expected learning outcomes.

Discipline-specific goals can be sourced from program guidelines and department resources. In the case of courses redesigned into CUREs, the learning goals articulated for the non-CURE format of the class should also be considered. Learning goals adapted to CUREs, particularly learning goals that incorporate technical skills, can be found in the CURE literature (e.g., Connor et al., 2022; Hanauer et al., 2017; Mishra et al., 2022; **Table 4**) and select dedicated publications (e.g., Irby et al., 2018). An additional source of learning goals may be found in the discipline-based education research literature, particularly concept inventories. Concept inventories and concept assessments, more than lists of discipline-specific learning goals, enable the evaluation of student learning through validated sets of multiple-choice questions (Libarkin, 2008). There exists concept inventories for a large number of disciplines and topics, particularly in STEM (Libarkin, 2008; **Table 6**); there are also databases of learning objectives for some disciplines (e.g., Bioliteracy; Chemistry, Life, the Universe, and Everything).

Discipline/Topic	Reference
Biology	
Natural selection	Anderson et al., 2002
Homeostasis	McFarland et al., 2017
Meiosis	Kalas et al., 2013
Genetics	Smith et al., 2008
Molecular biology	Couch et al., 2015
Genetic drift	Price et al., 2014
Evo-devo	Hiatt et al., 2013; Perez et al., 2013
Dominance in genetics	Abraham et al., 2014
Microbiology	Paustian et al., 2017
Microbiology for health sciences	Seitz et al., 2017
Biological experimental design	Deane et al., 2014
Population dynamics	Q4B Project
Speciation	Q4B Project
Translation and Transcription	Q4B Project
Statistical reasoning in biology	Deane et al., 2016
Host Pathogen Interactions	Marbach-Ad et al., 2009
Earth Sciences	
Meteorology	Davenport & French, 2020
Oceanography	Arthurs et al., 2015
Mineralogy	Scribner & Harris, 2020
Climate change	Jarrett et al., 2012
Physics	
Thermal physics	Yeo & Zadnik, 2001
Newtonian gravity	Williamson et al., 2013
Psychology	
Research methods and statistics	Veilleux & Chapman, 2017
Astronomy	
Planet formation	Simon et al., 2019
Chemistry	
Redox	Brandriet & Bretz, 2014
Enzyme-substrate interactions	Bretz & Linenberger, 2012

Table 6. Examples of published concept inventories

Learning outcomes can be drawn from program requirements and general education curriculum expectations. At the Ohio State University, competencies and skills that are not discipline-specific are articulated in the expected learning outcomes of the foundations, themes, integrative practices, and embedded literacies of the general education curriculum (Ohio State GE Program). Like at many other institutions, additional expected learning outcomes exist for the Honors program (Honors Program Goals).

Foundations courses enable students to gain a well-rounded education across academic disciplines and modes of inquiry. At Ohio State, students are required to take a course in each of seven fields of study (GE Program Structure), each associated with five to eight ELOs. Theme courses have their own sets of ELOs based on the theme they fall under. Theme courses are particularly appropriate for CUREs because the implementation of a CURE in a theme course enables it to qualify as a high impact practice course. The integrative practice inventory itself dictates ten expectations for these courses; they should be carefully considered at the time of design. Within a major, courses at the introductory or upper level can be valuable contexts to develop CUREs. Courses within the major are driven in part by the ELOs of the embedded literacies. Honors courses should satisfy ELOs from the Honors program.

Instructors should select learning outcomes from the appropriate list based on the nature of the course being developed (see question 1 starting on p. 51). Based on the discipline of the course, a subset of the ELOS for the foundation courses, theme courses, or embedded literacies are relevant. Once the ELOs for the course have been selected, they should be translated to specific outcomes for the CURE being developed. Specific learning outcomes should be center ed around students. All statements should start with “Student will be able to …” This statement should be followed by an action verb appropriate for the goal (see Verb wheel for examples). Each CURE outcome should also be associated with specific assignments that will enable the instructor to assess whether students have met the desired course outcome; in application of the backward-design approach (see p. 32). Examples of such alignment efforts are shown below for select learning outcomes. Instructors developing their own CUREs can contact the Office of Academic Enrichment (https://osas.osu.edu/oae/) as well as the Office of Undergraduate Research and Creative Inquiry (https://ugresearch.osu.edu/) for assistance in translating the learning outcomes of university-wide programs into course-specific ELOs. Best practices in writing and using learning goals are reviewed in Orr et al. (2022) at http://lse.ascb.org/learning-objectives/.

For each of the four curricular context identified above for Ohio State, **tables 7-10** show the alignment between select ELOs, their translation for different example CURES in different disciplinary contexts (STEM, social sciences, and humanities or any), and a proposed assignment enabling the evaluation of the ELO. The assessment of learning outcome competency is critical and should be integrated in the design process of the course (see Mishra et al., 2022 for an example).

Examples of ELO alignments for Foundation courses

Select Expected Learning Outcomes for Foundation courses associated with examples of translation to specific CUREs and proposed assessments

Expected Learning Outcome	Translation to specific CUREs	Proposed assessment
Generate ideas and informed responses incorporating diverse perspectives and information from a range of sources, as appropriate to the communication situation	Compare results of analyses to published findings	Discussion section of the manuscript will be assessed for explicit comparisons with the results of prior studies on different organisms
	Integrate perspectives from different actors to explain social interactions	Compare and analyze the transcripts of the interviews coding for commonalities

Expected Learning Outcome	Translation to specific CUREs	Proposed assessment
	Organize raw narrative data into usable research data	Develop an indexing system (e.g., titles and keywords) for organizing oral histories into a searchable database
Draw appropriate inferences from data based on quantitative analysis and/or logical reasoning	Interpret results of statistical analyses of phenotypic data in light of ecology	Results and discussion sections of the manuscript will be assessed for match between statistical test results, conclusions about significance, and interpretation in terms of locomotion
	Correlate quantitative demographic data and questionnaire responses to test specific hypotheses	Graphical analysis of the relationship between anthropometric data and views of others' bodies
	Analyze word-frequency data to identify linguistic changes	Quantitatively identify important language change events and their relationship with political, historical, and sociological phenomena
Analyze and interpret significant works of visual, spatial, literary and/or performing arts and design	Integrate historical context and iconographic analysis to critically assess the representation of past events	Write a critical analysis of a colonial art painting from the MET collection
Use historical sources and methods to construct an integrated perspective on at least one historical period, event or idea that influences human perceptions, beliefs, and behaviors	Combine the writings of different authors representing opposite interest groups to understand historical upheavals	Synthesize into an essay the works of revolutionary and anti-revolutionary proponents to qualify the atmosphere prior to the July revolution in France in the early 19th century
Employ the processes of science through exploration, discovery, and collaboration to interact directly with the natural world when feasible, using appropriate tools, models, and analysis of data	As a group, assemble a database from field specimens that will enable novel analyses	Collaboratively build a thorough dataset of qualitative and quantitative morphological information from wild plants for the Ohio State Marion prairie
Critically evaluate and responsibly use information from the social and behavioral sciences	Assess the value of published data and the context in which these were collected	Analyze a time-series of census data and rigorously determine the possible effects of biases on the data
Explain how categories including race, ethnic and gender diversity continue to function within complex systems of power to impact individual lived experiences and broader societal issues	Associate personal narratives with patterns in socioeconomic data	Review journalistic articles presenting personal stories of people representing race, ethnic and gender diversity and link them with quantitative data from the social science literature

Table 7. Examples of ELO alignments for Foundation courses

Examples of ELO alignments for Theme courses

Select Expected Learning Outcomes for Theme courses associated with examples of translation to specific CUREs and proposed assessments

Expected Learning Outcome	Translation to specific CUREs	Proposed assessment
Identify, reflect on and apply the knowledge, skills and dispositions required for intercultural competence as a global citizen	Evaluate the component skills of intercultural competence	Reflection on personal competency in the components of intercultural identity
Demonstrate a developing sense of self as a learner through reflection, self-assessment and creative work, building on prior experiences to respond to new and challenging contexts	Assess changes in metacognition over the course of the CURE	Reflection workbook activities on information literacy
Engage in an advanced, in-depth, scholarly exploration of the topic or idea of sustainability	Summarize the state of the research on the management of natural resources	Collaborative literature review activity summarizing papers of the students' choosing on the management of natural energy resources
	Analyze the roles of different stakeholders in the adoption of sustainability policies and commitments worldwide	Critical essay of the roles of political and economic stakeholders in a decision made at the COP26
	Synthesize the historical perspective of the concept of sustainability	Write an essay on the historical developments that have led to the modern concept of sustainable development
Explore and analyze health and well-being from theoretical, socio-economic, scientific, historical, cultural, technological, policy and/or personal perspectives	Synthesize health outcome data from multiple fields of study	Integrate data from the medical literature and experimental work in model animals to inform the relationship between pollution and chronic health conditions
	Evaluate how mental attitudes affects health efforts	Test the hypothesis that positive mental attitudes lead to significantly greater engagement in healthy behaviors
	Summarize historical data on population health	Compare health metrics in England between Tudor and Victorian times

Table 8. Examples of ELO alignments for Theme courses

Examples of ELO alignments for Embedded literacies

Select Expected Learning Outcomes for Embedded literacies associated with examples of translation to specific CUREs and proposed assessments

Expected Learning Outcome	Translation to specific CUREs	Proposed assessment
Apply methods needed to analyze and critically evaluate statistical arguments	Re-analyze published data to evaluate the validity of proposed paradigms	Graphic of the distributions of isotope masses and associated summary statistics for previously analyzed supernovae
		Critical evaluation of published quantitative data of three select psychoanalysis studies
		Use network analysis to investigate the World Religions Paradigm
Interpret the results of qualitative data analysis to answer research question(s)	Use visual analysis of rock samples to determine lithology in the field	Lithological characterization of rock units of geological map
	Employ questionnaires to investigate major cultural anthropology questions	Survey immigrant populations in the Columbus area to shed light on the processes that help groups of people maintain their culture
	Explain the role of theatre reviews in the dramatic arts	Synthesize reviews of theatrical performances to characterize the status of theatre through time
Develop scholarly, creative, or professional products that are meaningful to them and their audience	Cooperatively develop a publication-quality manuscript in a given journal format	Manuscript accompanied by self and peer-reflection of contribution to the final product
Recognize how technologies emerge and change	Articulate the significance of a recent technological advancement for future research in chemical engineering	Group Pecha-Kucha presentations of case studies involving micro computed tomography in chemical engineering
	Appreciate the importance of technological exchanges across the world	Create a map of technological advancements that led to the development of the computer mouse
	Explain the role of historical advancements in technology on human societies	Essay on the consequences of the invention of the printing press on the dissemination of information in Europe during the 15th and 16th centuries

Table 9. Examples of ELO alignments for Embedded Literacies

Examples of ELO alignments for Honors courses

Select Expected Learning Outcomes for Honors courses associated with examples of translation to specific CUREs and proposed assessments

Expected Learning Outcome	Translation to specific CUREs	Proposed assessment
Reflect on ways in which their learning furthered their aspirations	Identify professional skills improved through the CURE	Pre/post self-efficacy and aspirations survey (e.g., Martin et al., 2021)
Identify, assess, and compare how scholars from a diversity of perspectives in different fields and disciplines approach their most challenging problems	Integrate knowledge from different subfields relevant to your research project	Contrast the strengths and weaknesses of molecular and morphological approaches in resolving phylogenetic relationships within Mammalia
	Contrast the policy solutions proposed by academics from different leanings	Evaluate the differences between the policies suggested by economists from various schools of thought in response to the Great Recession
	Undertake a comparative analysis of a topic treated by different writing modes	Contrast a non-fiction graphic novel and a biography or historical/journalistic book discussing the same topic
Take on a variety of roles within groups	Use appropriate language and communication skills to fulfill a given role within a group	Effectively engage in group discussion in the different capacities identified by the instructor
Communicate using modalities that are effective and inclusive relative to the intended audience	Model inclusive interactions during group activities	Self and peer-evaluations of group work
Articulate what success looks like for them in both personal and professional contexts	Develop three personalized learning goals for the semester	Self-assess progress on these goals and participate in an end of the semester conference
Demonstrate a growth mindset to integrate new information and ways of thinking	Evaluate self-efficacy and acquisition of new skills	Reflect on self-efficacy and acquisition of new skills pre and post CURE

Table 10. Examples of ELO alignments for Honors courses

E) Structuring your CURE

Jonathan J.-M. Calède

Designing a CURE is a complex endeavor that requires coordination with staff members, administration, and other instructors. Fortunately, there is now an extensive literature on CUREs that enables the identification of best practices and recommendations for an effective experience. The elements identified in **Table 11** should be revisited throughout the process of designing a CURE.

Topic and Focus	Topic is appealing
	Research is publishable / significant to others
	Project ownership is encouraged
	Instructor leverages their expertise to foster high-level research interactions
Research Design	Research only requires minimal background and is conceptually simple
	Appropriate technical expertise is required to collect data, especially in initial stages
	Study system involves enough variables to allow for a variety of questions
	A common database of variables enables authentic research and peer-discussions
Pedagogical Design	Project information is regularly and appropriately provided
	Scaffolding of research activities guides students through the process
	There are frequent deadlines
	Multiple achievement milestones are built into the schedule of activities
	Everyday strategies implemented in the classroom explicitly align with ELOs
Communication	Communication between student and instructors as well as among students
	The frequent feedback structure incorporates peer-reviews and leads to revisions
	Specific guidelines and expectations for communication and inclusivity are explained
Reflection	The course structure provides opportunities for reflection on the research process as well as the specific project
	The course structure includes a reflection on personal learning and gains
	Course assessment mirrors authentic scholarly communication

Table 11. Elements of a successful CURE (modified from Dolan, 2016; Hanauer et al., 2022; Kortz & van der Hoeven Kraft, 2016; Wiley & Stover, 2014; Turner et al., 2021; Kloser et al., 2011; Hatfull et al. 2006)

The goal of this section is to use the framing questions presented above (p. 33-34) to guide prospective CURE instructors through the development of the research experience. These questions are not just prompts for reflection; together, they also form a worksheet for the design process.

- For each question, the "Overview" provides an overview of the critical design elements that are identified and developed in this section of the worksheet.
- The "starts here" units guide you through these elements of design. Although the pedagogy-centered issues are presented before the research-centered ones, they are better thought of as parallel tracks.
- The "research and education together" section combines the elements of design from both aspects of the CURE. This is an opportunity to identify and resolve conflicts.
- The "boxes" help remind instructors of the important elements of the question and critical points of contact across (and beyond) campus.

1. How will the CURE be integrated into the curriculum?

Overview: Despite the desire to design the CURE solely around learning goals and research questions, reality requires first the consideration of the place of the CURE in the students' slate of courses. This section enables you to determine:

1. The audience of your course
2. The level of preparation and prior knowledge you can expect from students
3. The duration of the experience
4. The integration of program requirements into the course
5. The scope and intent of the research

Education starts here: Many of the questions below are best answered in communication with the chair of the undergraduate curriculum committee of your department or unit.

- Will the CURE be a new course or integrated into an existing course?
- If developing a new course:
  - Will this course be aimed at majors or non-majors?
  - Will this course be an upper-level class with prerequisites or open to freshmen and sophomores?
  - Will this course become a requirement for any minor or certificate?
  - How many course credits will the CURE represent? Are there constraints on this number? How many hours of contact hours does this correspond to? How will they be divided between lectures, recitations, and labs? What is the total time commitment you can expect from students including homework?
  - Will this course be taught as a summer course? A half-semester course? Through two courses over an entire academic year?
  - What is the expected enrollment for the course?
  - Will this course become a requirement for other courses or be an elective?
    - If becoming a requirement:
      - What is the knowledge base and skillset that students need to master to continue with the next course?
    - If an elective:
      - What are the goals and expected learning outcomes of electives in your department, college, or unit? More broadly, what are expectations in your field and related professional avenues?
- If revising an existing course:
  - What is the place of this course in the curriculum? The major? Minors or certificates?
  - Will the CURE be replacing traditional labs or are you also integrating the lectures and/or recitations into the experience?
  - What are the expected learning outcomes of the current version of the course? Are they being revised?
  - What is the current enrollment in the course?
  - How does the course fit into the curriculum and how does it prepare students for the next steps in their education and career?

Scholarship starts here: When developing a researcher-driven CURE, this element of design is led by the researcher and his team. When developing a researcher-independent CURE, this element of design is driven by the CURE program goals that you are joining.

- What discipline/field of research does the research topic fall under?
- Is the research project novel? What is already known about the possible outcome(s) of the research?
- Are you planning on focusing on a single question/goal or will the class tackle several research questions?
- Is this research hypothesis-driven or exploratory?
- What is the current stage of the research?
- Who are the stakeholders of the research that should be involved? Any collaborators?
- How does the research align with the topic, scope, and knowledge goals of the class (particularly if revising a pre-existing course)?

Research and Education together: Any incompatibility between research and education goals at this point of the development process, including incompatibilities between research goals and class enrollment or number of contact hours, may be fatal. Although revisions to the focus of the research or learning goals of the course are sometimes possible without compromising the student experience and learning, the nature of the course and the scope of the research must come together for the CURE to be possible.

Box 1: Important points

- Contact the chair of the undergraduate curriculum committee of your department or unit.
- Verify the guidelines and requirements of the course approval process when proposing new courses.
- Engage colleagues (staff or faculty) involved in the implementation of existing courses if you are revising them.
- Identify the relevant expected learning outcomes of the department, program, major, minor, or certificate.
- Do not forget the research stakeholders. How can postdocs, graduate students, and undergraduate students in your lab contribute to and benefit from the CURE?

2. To what extent will students have intellectual responsibility and ownership of the research?

Overview: Research shows that student ownership of the research project undertaken as part of the CURE positively impacts their experience of the course and their learning gains (Hanauer et al., 2012; Harrison et al., 2011; Hatfull et al., 2006). Project ownership is a central component of the role of CURE instructors (Hanauer et al., 2022). Different degrees of project ownership may be possible depending on the scope, depth, and conditions of the CURE. They are intimately associated with the level of inquiry that students engage in.

Several levels of inquiry have been defined in the context of laboratory courses (Buck et al., 2008) and revised to fit the framework of CUREs (Brownell & Kloser, 2015). Because CUREs require the investigation of novel questions with no-known outcomes and the communication of the research results, they may only fall under some of these categories (**Table 12**). The degree of responsibility of the students in the design of the research project is associated with the degree of inquiry desired by the instructor and should be guided by the expected learning outcomes of the course. Together, they will open a range of opportunities for the student's ownership of their research.

Characteristic	Structured Inquiry	Guided Inquiry	Open Inquiry	Authentic Inquiry
Problem/Question	Provided	Provided	Provided	Not provided
Theory/Background	Provided	Provided	Not provided	Not provided
Procedures/Design	Provided	Not provided	Not provided	Not provided
Results analysis	Not provided	Not provided	Not provided	Not provided
Communication	Not provided	Not provided	Not provided	Not provided
Conclusions	Not provided	Not provided	Not provided	Not provided
Known answer?	Yes/No	Yes/No	No	No

Table 12. Levels of inquiry in undergraduate courses (modified from Buck et al., 2008 and Brownell & Kloser, 2015). It is entirely possible to imagine hybrid situations in which for example only parts of the research protocol are provided to students.

This section enables you to determine:

1. The level of inquiry you are aiming for in the CURE.
2. The choice between students exploring researcher-chosen questions/hypotheses and student-driven questions/hypotheses
3. The amount of input students will have on the design of the research project
4. The ability of students to communicate the research upon its completion

Education starts here: The ELOs for the course are critical in determining everything from the nature of the research activities included in the experience to the scope of the research questions. They should be referenced throughout the course design process.

- What level of inquiry is dictated by the ELOs?
  - Are students expected/able to develop original questions directly from observations? Will students be choosing from a range of predetermined questions and topics?
  - How will you foster peer discussions and class-wide activities if several questions/hypotheses will be studied by different groups or individuals?
  - Will you be giving students an introduction to the existing knowledge on their research topic or are they expected to collect this information directly from the published literature? If the latter, how will you guide this work and/or assess it?
  - Are students allowed/capable of collecting their own data based on permitting requirements (e.g., IRB) and ethical best practices?
  - Will appropriate analytical procedures be suggested to students or should they propose particular approaches based on their reading of the literature?
- Is the level of inquiry selected compatible with the amount of time given to students?
- Is the level of inquiry selected compatible with the expertise of the students?
  - What is the efficacy of students with reading the primary literature? Will you be providing training on this topic?
  - What is the familiarity of students with the field of research chosen for the CURE? How much background information is necessary to be able to comfortably initiate the research process?
  - Can you expect the students to understand and implement the analytical procedures likely to be involved in the research? Will you be providing the necessary knowledge as part of the class?
  - What are the computational skills of the students? Does the analytical protocol require coding skills or can it be implemented in a software program with a GUI?
  - In what prior classes or context would the students have acquired the required expertise/competence for the CURE?
- If students will explore their own questions:
  - Will you be placing constraints of the breadth and depth of the questions chosen by students?
    - What is too narrow a question?
    - What is too broad a question?
    - Can you mentor the diversity of questions and datasets entailed by this structure?
    - Are there topics that could be emotionally difficult for some students? Will these students have the opportunity to work on a different projects or will these projects simply not be validated by the instructor?
  - How will students select their hypothesis/question from all possible options?
    - What is the process by which a student question/hypothesis is approved for the CURE?
    - Can a mock panel (for example composed of the instructor, graduate student researchers, and colleagues) review student proposals?

- If students will be working in groups, how will you balance the group structure with individual questions?
- If students will be exploring researcher-chosen questions:
  - Will the questions be prescriptive, or will there be an opportunity for students to refine their question, narrow it, or define a question from a broad topic?
  - Is the question chosen compatible with the time frame of the CURE and the skill level of the students?

Scholarship starts here: It is important to recognize that there are constraints on scholarship stemming from regulatory and ethical requirements. Additionally, and no less important to the success of the CURE, its place within the broader research project it is part of, may drive the nature of the deliverable, the breadth and depth of the work, and the level of prescription imposed by the instructor(s).

- What are the existing constraints on the project from other stakeholders (e.g., other lab members, external collaborators, and national programs or networks)?
- What are the existing constraints on the project from existing data, commitments to funding agencies, museum collections, permitting agencies, university rules (including EHS and IRB), formal and informal agreements, and ethical concerns?
- How do these different types of constraints narrow the nature, breadth, and depth, of the questions that can be investigated by the students? How about the analytical protocols?
- If the CURE is researcher-driven:
  - What question are you interested in exploring?
  - What question/topics are you comfortable/uncomfortable mentoring?
  - How much of the research process are you comfortable leaving up to the students?
  - What is the deliverable you are hoping for upon completion of the CURE?
  - How critical is the success of the research (i.e. the obtention of the deliverable defined above) to you?
  - How much time will you be dedicating to the CURE outside of contact hours?
  - Is the research goal of the CURE compatible with the time and means given to students?
- Does the CURE represent a standalone project publishable upon its completion or a portion of a bigger project? How does this impact the format of research communication upon completion of the CURE (i.e. does the end product represent a manuscript? A poster presentation? An element of a grant proposal?)

Research and Education together: Legal, ethical, and professional requirements dictated by the field of research, necessary resources for the project, and the research program context of the CURE are the starting point for the level of inquiry that is possible for the CURE. However, there are several existing strategies to increase student engagement and ownership of the project that should be considered:

- Hanauer et al. (2022) presented a model of CURE mentoring that includes a project ownership strategy centered around fostering personal responsibility starting with teaching a scientific protocol and includes promoting research ethics, facilitating peer collaboration, encouraging independence, encouraging engagement and enthusiasm, creating opportunities for presentations, and fostering future educational and career opportunities.
- Students who collect their own data are more invested in the research project and display a greater research identity than students working with pre-existing datasets they did not collect (Cooper et al., 2020a).
- Undergraduate students can work with graduate students or undergraduate researchers who have benefited from mentored research experience or previous iterations of the CURE to contribute to the research questions and methods within a framework (Hanauer et al., 2012).
- Emphasizing to students the significance of the research to the broader community of scholars in the field is important (Hanauer et al., 2012). Research shows that broadly relevant novel research leads to higher ownership of the project by students (Cooper et al., 2019).
- It is sometimes possible to pursue research questions relevant to the community or the students themselves (e.g., Malotky et al., 2020; Silvestri, 2018; Valliere 2022a). Students enjoy the opportunity to choose their own research topic (Amir et al., 2022) and those who are given opportunities to investigate questions relevant to themselves or their community show increased ownership of the research (Hanauer et al., 2012).
- Incorporating meetings that mimic research group meetings in the discipline and poster presentations can promote project ownership (Satusky et al., 2022).
- A CURE design that creates intellectual challenge and encourages problem solving deepens the engagement of students and the significance of the research experience (Hanauer et al., 2012). Challenges and iterations are critical to increasing the ability of students to navigate research obstacles (Gin et al., 2018; Light et al., 2020). In fact, students report valuing the ability to learn from their mistakes (Harrison et al., 2011) and view challenges and iterations as more representative of a real research experience (Goodwin et al., 2021). CUREs should deliberately incorporate iteration and discussions about the importance of iteration in research as part of student development (Light et al., 2020). Iteration enables the development of adaptive strategies by students that benefit them beyond the course (Cooper et al., 2022). A review of the best practices to engage students in problem solving is provided by Frey et al. (2022) and at https://lse.ascb.org/evidence-based-teaching-guides/problem-solving/.
- The CURE should be challenging without being overwhelming; this level of difficulty, paired with instructor support, has previously be showed to foster motivation (Dolan, 2016).
- If students will be developing their own questions, they should be guided through the process of identifying their own research question including the following critical issues:
  - What questions have already been asked?
  - What is the existing knowledge on the topic?
  - What questions have not yet been answered or even asked?
  - What are the gaps in the conversation?
  - What questions are relevant to the field of research?
- An activity leading students through the process of identifying scholarly significant questions can be followed up with an activity helping them narrow big research questions into manageable CURE projects (see [**Activity 2**]).

Box 2: Important points

- Students should tackle questions and test hypotheses relevant to the broader (research) community.
- Students should be allowed to make decisions throughout the design and implementation of the research protocols.
- Instructors should rethink their place in the classroom to that of mentors and not merely supervisors.
- There should be multiple opportunities for students to develop their own hypotheses and defend them with evidence from the literature and their own work.

3. Which components of the research process will be integrated into the cure?

Overview: The traditional view of the scientific method involves a series of steps starting with observation and the formulation of a hypothesis followed by the test of this hypothesis and the dissemination of the findings (Voit, 2019). In the social sciences, qualitative, quantitative, and mixed approaches may be adopted to probe questions pertaining to human beings and their societies, but the overall process of inquiry remains the same (Creswell, 2014). The nature of the data collection process and data analyses differs in the humanities, but the scholarly endeavor is still the "evidence-based exploration of a question or hypothesis that is important to those in the discipline in which the work is being done" (University of Washington English Department). Thus, across disciplines, the research process requires the identification of an appropriate research question, whether it be from direct observation or a review of the primary literature, the collection of data, their analysis, the interpretation of outputs, and the communication of the knowledge gained from the research. It is critical to identify which of these elements of the research process will be included in the CURE while considering the constraints of the undergraduate classroom. This section enables you to determine:

1. Your approach in defining research to students and framing their experience as scholars
2. The background information that needs to be provided to the students
3. The role of students in the data collection process
4. The engagement of students with the primary literature
5. The role of students in the data analysis process
6. The role of iteration and impact of failure on the CURE

Education starts here:

- How would your students define "research"?

- What does the research process look like to them?
- What is their understanding of the concept of “Research as Inquiry?”
- What is your understanding of the students’ view of research?

- What information is necessary to understand the basis of the question(s) the students will investigate?
  - Are there recently published reviews of the field/topics that students will be researching? Are there other sources of information that can help draw students into the literature (including secondary and tertiary literature, videos, and journalistic writings)?
  - Are there important conflicts or controversies in the field that students should know about? What is the current paradigm and are we on the edge of a new one?
  - Will you provide a brief introduction to the study system in class drawing upon your knowledge, past iterations of the CURE, and ongoing research on campus (and beyond)?
- Will students be collecting the entire dataset they will analyze? Alternatively, will they be working from pre-existing datasets in whole or in part?
  - Are there pilot or example datasets that enable students to visualize their objective while collecting most of the data they will analyze?
  - Will specific data collection protocols be enforced by permitting requirements (e.g., IRB) or ethical best practices?
  - Are there professional, legal, or ethical training requirements for students to collect data? Consult and analyze data?
    - How can you incorporate responsible and ethical conduct of research training into the course to satisfy needs and train reflective and responsible professionals?
- Will students be required to read certain publications or a minimum number of freely chosen publications? At what stages of the research process will engagement with the literature be suggested/enforced?
  - How will you guide students towards relevant publications and/or vet their choice of readings?
  - How will you make sure that students engage with a variety of sources and read the work of multiple authors representing different points of view?
  - Students should be prompted to reflect upon questions like: (1) How does the source contribute to the scholarly conversation? (2) How do others in the field perceive the value of the source? (3) How does the source guide or support my work?
- Will specific analyses be required of students in the form of a detailed protocol or as a list of milestones? If not, how will students choose the analyses they will use? How will they be guided through this process?
  - What is the role of published studies as models for the students’ work?
- Will students identify analyses that will be run by scholars with greater computational skills or are they able to run some or all of these analyses themselves?
- How would experimental failure or faulty data collection impact the learning process?
  - Are there backup datasets?
  - Are there alternative protocols and analyses?
  - What is the relevance/significance of null results or a failed protocol to the scholarly community?
- Is there enough time in the CURE to implement alternative protocols? Revise the data collection? Add to the dataset? Repeat an analysis?

Scholarship starts here:

- Is there a need for a formal review of the field of study the students will engage with?
  - Can the CURE be the impetus to publish a peer-reviewed review of a topic?
  - Would a more accessible summary benefit audiences beyond the CURE including undergraduates starting mentored research experiences in the lab or the general public?
- Is the CURE an opportunity for the instructor/researcher to engage or reengage with specific aspects of the primary literature?
- What is an appropriate sample (including quantity and quality parameters) to address the question of the CURE?
- Will student-collected data represent the entire set of relevant data for this project or are there other elements of the research project they will not engage in?
- Does the use of preexisting data dictate a specific data collection protocol to enable comparisons/integrations?
- What work is necessary ahead of the CURE to make data collection by students possible?
- What checks and balances will be implemented to validate the quality of the student-collected data?
  - Do those checks and balances involve peer-review? Instructor review? Outside researcher review?
- Are there analyses that are standard/expected in the field?
  - Are some of these analyses computationally too demanding for the CURE?
  - Do some of these analyses require coding or statistical training beyond what can reasonably be expected of students?
- Are there critical failure points in the research protocol? Are there data or analyses whose failure would hobble or interrupt research progress?
- What would be the consequence of failure (of a specific analysis, experiment, or data collection) on proposal submissions, publication completions, and professional advancements (including promotion and tenure)?
- What is the role of the CURE in the development of the research program/project(s) of the researcher/instructors/collaborators/graduate students/mentored undergraduate researchers/ etc.?

Research and Education together:

- Do not underestimate the importance of framing the entire CURE experience with a conversation about the nature of research. Students may not always recognize research as information creation and inquiry. “Research as Inquiry” is a core concept underlying the practice of research. Defining research as inquiry means explaining to students that research is an open-ended and messy exploration process focused on information gaps, unanswered questions or problems, involving multiple sources and the interpretation of information, often with no clear right answer, leading to new questions, generating ambiguity, and requiring an open mind, persistence, and flexibility (e.g., [**Activity 3**]). Research shows that engaging in this type of conversation about the nature of research can lead to higher student outcomes (Lopatto et al., 2022).
- CUREs are designed as authentic engagements in research and as such should incorporate responsible conduct of research (RCR) training to form conscientious professionals (Diaz-Martinez et al., 2021). Many institutions and organizations require ethical conduct of research training. At Ohio State, RCR training is provided through the Collaborative Institutional Training Institute (CITI) course, which can be assigned as homework.
- Discipline-specific best practices in ethical conduct of research should also be implemented as part of the CURE (e.g., Hills and Light 2022). They can be integrated in the grading scheme for the course.
- Consider using mind-mapping exercises to help students identify the existing knowledge of the topic of the CURE and the current debates and controversies in the field of study [**Activity 4**].
- Established databases provide sources of data that can be used in CUREs (e.g., Gastreich, 2020). Those include museum databases, citizen science databases, and professional databases (**Table 13**).

Examples of databases that can be leveraged for research projects in CUREs

Research databases across disciplines including Art History, Biology, Earth Sciences, Machine Learning, Physics and Astronomy, as well as Political and Social Sciences

Name	URL
Art History	
Index of Medieval Art	https://theindex.princeton.edu/
Wax collection of Islamic art	https://minicomp.github.io/wax/collection/
Biology	
AntMaps	https://antmaps.org/
AntWeb	https://www.antweb.org/
Arctos	https://arctos.database.museum/

Name	URL
CalFlora	https://www.calflora.org
CalPhotos	https://calphotos.berkeley.edu/
COSMIC	https://cancer.sanger.ac.uk/cosmic
Diatoms of North America	https://diatoms.org/
eBird	https://ebird.org/home
featherbase	https://www.featherbase.info/en/home
iNaturalist	https://www.inaturalist.org/
Madagascar Terrestrial Camera Survey	doi/10.1002/ecy.3687
Movebank	https://www.movebank.org/cms/movebank-main
NEON	https://www.neonscience.org/
NEOTOMA	https://www.neotomadb.org/
NOW	https://nowdatabase.org/
Paleobiology Database	https://paleobiodb.org/
Protein Data Bank	https://www.rcsb.org/
VertNet	http://vertnet.org/
Earth Sciences	
Macrostrat	https://macrostrat.org/
Soil grids	https://soilgrids.org/
WorldClim	https://www.worldclim.org/
USGS Current Water Data	https://waterdata.usgs.gov/nwis/rt
LAGOS	https://lagoslakes.org
EarthData	https://search.earthdata.nasa.gov/search
Stone Lab Algal and Water Quality Laboratory	https://ohioseagrant.osu.edu/research/live/water
Mauna Loa Trends in Atmospheric CO2	https://gml.noaa.gov/ccgg/trends/data.html
NASA GISS Surface Temperature Analysis	https://data.giss.nasa.gov/gistemp/station_data_v4_globe/
North Temperate Lakes LTER	https://lter.limnology.wisc.edu/data
Machine Learning	
Machine Learning Repository	https://archive.ics.uci.edu/ml/index.php
Physics and Astronomy	
SIMBAD	http://simbad.u-strasbg.fr/simbad/
Astrophysics Data System	https://ui.adsabs.harvard.edu/
Political and Social Sciences	
ICPSR	https://www.icpsr.umich.edu/web/pages/
Various	
Smithsonian Open Access	https://www.si.edu/openaccess
Hathi Trust Digital Library	https://www.hathitrust.org/datasets
GeoPlatform	https://www.geoplatform.gov/
Omeka Project Directory	https://omeka.org/classic/directory/

Table 13. Examples of databases that can be leveraged for research projects in CUREs.

- Iterative CUREs enable students to access data from previous years and other lab groups and as such the opportunity to analyze realistic sample sizes that may be difficult or impossible to gather in the time of a CURE (Kloser et al., 2011; see Satusky et al., 2022 and Sun et al., 2020 for examples). The assembly of large datasets also enables long-term outlooks (e.g., Potter et al., 2009).
- It is important to define "Scholarships as Conversation" for students. Scholars and researchers engage in ongoing conversations in which new ideas and research findings are continually being discussed (ACRL) (e.g., [**Activities 5-6**]).
    - Being part of the scholarly conversation is an expectation of academic research and is integral to research and therefore of a CURE.
    - The scholarly conversation takes place in the peer-review literature, including books and journal articles.
    - Scholarly conversations also take place at conferences.
    - The classroom is an environment for scholarly conversations.
- There is an extensive literature on teaching the primary literature. There also exists multiple activities and exercises that enable instructors to introduce the primary literature to undergraduate students. Examples can be found in [**Activity 7**], Beck (2019), Hammons (2021), Howard et al. (2021), Chen (2018), Carson & Miller (2013), Hartman et al. (2017), Mitra & Wagner (2021), and Smith (2001) as well as the C.R.E.A.T.E. strategy (https://teachcreate.org/roadmaps/) and the science education resource center at Carleton College (Egger; Mogk). The collaborative reading and annotating of articles from the primary literature by students can also be helpful (Cafferty, 2021).
- Providing guidelines and help on exploring the primary literature through formal (or informal) bibliography exercises (e.g., [**Activity 8**]) helps students find additional resources on their own (e.g., [**Activity 9**]).
- Encourage students to identify the consensus that have developed, but also competing perspectives and approaches, and how new voices and evidence emerge (e.g., [**Activities 9-11**]).
- Do not overlook the importance of explaining to students how to evaluate and select sources as well as how to provide citations. Many students are not familiar with the process of searching for references (video).
- You should also engage students to reflect upon best practices of literature review, including:
    - Using existing research to guide one's own work (including on the issue of "tracing the scholarly conversation" [**Activity 9**]).

- Considering context when evaluating sources. One may not be able to understand the value of a particular piece of scholarship unless they consider the broader conversation (see [**Activity 10**]).
  - Providing attributions.
  - Needing to learn the "language" of the conversation before being able to fully participate.
  - Acknowledging that joining in the conversation confers both rights and responsibilities.
  - Recognizing that one is most likely entering into an ongoing conversation and not a finished one
- Are there freeware programs or programs with user-friendly GUIs that can be used by students to engage in the data analysis process without the need for extensive coding or other computationally difficult tasks (see Acuna et al., 2020 for an example; Zelaya et al., 2022 for an example in a CURE context)? An extensive collection of freeware programs for data visualization and analysis, network analysis, mapping, text analysis, etc. is available at https://guides.osu.edu/DH/digitalhumanities (see also **Table 14**).

Examples of freeware programs and programs with user-friendly GUIs that can help engage students in analyses

Select freeware programs enabling image analyses, statistical analyses, concept mapping, and visualization of data

Program name	URL	Type of analyses
ImageJ	https://imagej.net/software/fiji/	Image analyses
PAST	https://www.nhm.uio.no/english/research/infrastructure/past/	Statistical, time-series, and spatial analyses
jamovi	https://www.jamovi.org/	Statistical analyses
VUE	https://vue.tufts.edu/	Concept mapping
Google Jamboard	https://jamboard.google.com/	Concept mapping / Interactive whiteboard
SwissADME	http://www.swissadme.ch/	Drug discovery
UCSF Chimera	https://www.cgl.ucsf.edu/chimera/	Visualization and analysis of molecular structures

Table 14. Examples of freeware programs and programs with user-friendly GUIs

that can help engage students in analyses.

- It is important to reflect on the role of the CURE in the development of the research program of the researcher(s). CUREs can help enable the research program of a research group (see figure 2 in Sun et al., 2020), but careful planning and circumscribing of the role(s) of the CURE(s) are critical.

Box 3: Important points

- CUREs should integrate responsible and ethical conduct of research into the student training.
- Consider the use of published data, museum and research databases, data collected by collaborators, and data collected by previous iterations of CUREs.
- Set-up checks and balances for student-collected data.
- Formally training students in the reading and analysis of the primary literature is critical to their engagement with scholarship.
- Build room in the course structure for failure and iteration; they are important elements of the learning process.

4. How will research progress be balanced with student learning and development?

Overview: The strength of CUREs is their combination of research and teaching in a unique pedagogical experience for both students and researcher-instructors; it is also an important source of challenges. The ideal CURE would involve a steady progress of the research process that is accompanied by student gains and learning. These pedagogical goals can sometimes conflict with the research advancing at a necessary pace. Similarly, the required validation and repetition steps of many research protocols may not be needed to fulfill many learning goals. The structure of a course itself with the associated time-constraints can place limits on repeating experiments, increasing sample size, and even fully exploring a particular dataset or question.

This section enables you to determine:

1. The pace of the research process
2. The appropriateness of lessons on replication and statistical power to address the limitations of time in a CURE
3. The possibility to assign some analyses and research tasks to outside researchers
4. The role of peer-review as a combination of the scholarship review process and a pedagogical feedback
5. Inclusivity issues to consider alongside research progress and student development

Education starts here:

- How will you make explicit to students the pace of the CURE?
  - Will there be class meetings dedicated to specific data collection efforts, analyses, group brainstorming, peer-discussions, instructor conferences, etc.?
  - What activities will be assigned to students as homework?
  - What is the schedule of writing of the different sections of the deliverable?
  - Can you identify weekly goals for the project?
  - When will students be prompted to reflect upon their work? When will they be asked to discuss their work with peers or mentors? When will they be asked to formally review and critique their work or the work of others?
- What are the elements of the research process that will not be authentically included in the CURE?
  - Can any of those elements be modeled on smaller datasets by the students?
  - Can any of those elements be modelled on a different dataset of the same nature (published or not) as part of a class activity?
  - Can some complex analyses demanding high computational skills be demonstrated by the instructor or guests (e.g., a graduate student working on this type of analysis)?
  - Are there elements of the research that can be outsourced to the instructor or other researchers and merely explained or showed to students?
- What activities and requirements could represent obstacles to the engagement of all?
  - Are some specific lab tasks, field work, data collection protocols, etc. in conflict with student accommodations?
  - Are there activities that take place outside of the normal class time but need to happen on a specific day or time (e.g., attending to something in the lab, recording field observations at specific events)?
  - Is the literature that students need to consult to complete the work accessible and affordable to them?
  - Are there graphical representations of data or media used as part of the research that are not accessible to all?
  - Are the tools and support necessary for the success of all available on campus, including IT resources?
- How can you develop a supporting environment in which students explore research, manage their work, and fail safely?

Scholarship starts here:

- Are there specific deadlines associated with the research project including proposal deadlines, abstract deadlines, theses deadlines, manuscript deadlines, and commitments to collaborators or research students (graduate or undergraduate)?
  - Is the CURE work necessary for another step of the research process outside of the classroom?
  - Is the CURE work integrated with the research of a graduate student or mentored undergraduate student?
- Does the work that students will undertake in the CURE represent the work necessary to complete the research deliverable chosen? Alternatively, is the CURE research only one component of the deliverable?

- What analyses, if any, typically reported in supplementary information, are necessary for validation but cannot be fully incorporated in the CURE?
- Can the research questions and/or tasks be easily divided among groups or students?
- Can you collect the literature relevant to the project (or a representative subset of it) ahead of the CURE to share with students?
  - Can you share with students during the CURE examples of papers being published that are relevant to the research?
- What are important research concepts and analyses for the CURE that may require some explanation? Have you completed [**Activity 1**] yourself?

Research and Education together:

- The milestones of the CURE should mimic the different elements of the process of scholarly work from inception to presentation/publication. Thus, students should produce deliverables that are equivalent (at least in format) to the deliverables of professional scholars (e.g., Bakshi et al., 2016; Gastreich, 2020; Ramírez-Lugo et al., 2021).
- Peer-review and instructor feedback are models for the scholarly review process. Consider using the review framework of a prominent journal in your field or alternative rubrics (see [**Activities 12A and 12B**]) to guide students through this process and make peer-reviews valuable educational experiences.
- A consistent structure helps instructor and students keep track of the research and learning processes. One such structure is presented in [**Appendix 1**]. The structure is designed in eight different pedagogical steps that lead to a research deliverable as an integration of the two aspects of the CURE. Other examples of course structures are presented in the published literature (Bakshi et al., 2016; Bell, 2011; Bowling et al., 2016; Carr et al., 2018; Chen, 2018; Colabroy, 2011; DeHaven et al., 2022; Hekmat-Scafe et al., 2017; Mills et al., 2021; Murren et al., 2019; Ramírez-Lugo et al., 2021; Satusky et al., 2022; Sewall et al., 2020; Sweat et al., 2018; Thompson et al., 2016; Waddell et al., 2021; Wilczek et al., 2022; Zelaya et al., 2020).
- Dedicating class time to tutorials associated with formative assessments and/or discussions enables instructor(s) to verify that students are ready to undertake the homework and activities of the CURE.
- Dedicating class time to group meetings enables the instructor(s) to check in on all students/groups of students and address concerns and difficulties. Such meetings should be structured and scaffolded to promote constructive conversations. An example of the possible structure for one of these meetings is presented in [**Activity 13**].
- Field work can sometimes be implemented through asynchronous self-led field trips to overcome logistical obstacles (Washko, 2021).
- Lessons and interventions on specific topics, including replication and statistical power, may help students understand issues and concepts in research that cannot be undertaken in the CURE itself and overcome misconceptions (Schwartz et al., 2004).
- Conversations with staff members in the Office of Academic Enrichment, Disability Services, the University Libraries, and the Teaching and Learning Center (Drake Institute at the Ohio State University) can help instructors find compromise and solutions to the challenges of implementing research in the classroom.
- The TILT framework (Winkelmes et al., 2016) enables instructors to increase transparency in assignments. Including explicit links between assignments and tasks as part of the purpose of the assignment helps make explicit the progress through the scholarly process. An introduction to the TILT framework is available here.
- Transparency in the assignments can also be improved by working with students through “understanding assignments” activities. Such activities can be done individually or as a group and enable students to make sure they are meeting the expectations of the instructor without being impeded by jargon. It encourages students to think about what it is they are expected to do (examples of assignments available at UNC Chapel Hill).
- Universal Design for Learning (UDL) is a framework that helps design teaching products and structures that can be used by all without the need for accommodations. UDL is a proactive approach to adaptation in the classroom and benefits all regardless of their needs for specialized designs. UDL enables instructors to meet the diversity of their classroom. It includes several key components reviewed in Burgstahler, 2013. UDL includes the implementation of different media, additional scaffolds, technology, etc. (Burgstahler, 2009; Griful-Freixenet et al., 2017), but also expands to other issues. Consider how you can equitably support students through flexibility in deadlines, meetings, and schedules. This can include letting students set their own work hours or provide students the opportunity to attend remotely certain meetings. Learn what users need by conducting check-ins for group access needs during lab meetings and surveying users about the overall accessibility of the course experience. Consult with accommodation services staff to determine how you can optimize research and teaching practices and spaces and how to make them more inclusive for individuals with disabilities and specialized needs. Advocate on behalf of students with disabilities by communicating with accommodation services staff and encourage students to reach out to those services to get the support they need and deserve.
- Mentorship structure can involve peer-researchers and graduate students.
- Any mentorship structure can benefit from direct feedback from students. Communication is key to the success of the CURE.

Box 4: Important points

- Mini-workshops and group or class-wide activities on subsets of the dataset analyzed or published data comparable to those studied by the students can help students understand critical concepts that cannot be authentically explored in the CURE because of time.
- CURE students are integral members of the research team and can benefit from intellectual or data exchanges with other members of the research team, including graduate students, postdocs, and undergraduates in mentored research experiences.
- Consider presenting to students the trajectory of a research project from inception to publication, including the role of formal and informal peer-reviews of ideas, presentation conferences, and manuscripts.
- Consider how universal design for learning can be implemented in the CURE to facilitate the engagement of all students in the experience.
- Implement a mentorship model of the CURE students that fosters a safe environment for self-exploration and self-management (Palmer et al., 2015).

5. How will the research learning tasks be structured to foster students’ development as scholars?

Overview: The design of a CURE requires the incorporation of a carefully thought-through scaffold that enables students to engage with an intellectual challenge that is often beyond any they have encountered before. Well-constructed scaffolds are critical to the success of CUREs because they enable students to tackle appropriately challenging tasks (Lopatto et al., 2020) and support students through failures, leading to greater perseverance (Corwin et al., 2022) . Scaffolds are relevant to the design of the entire CURE, which should include a series of activities and assignments that guide students through the entire project (e.g., Bakshi et al., 2016; Delventhal & Steinhauer, 2020; Hills et al., 2020; Makarevitch et al., 2015; Peyton & Skorupa, 2021; Rennhack et al., 2020). They are also relevant to the design of individual assignments because they help guide students through the tasks of research. This section enables you to determine:

1. The appropriate framework for the CURE assignments
2. The structure of CURE tasks as individual or group assignments
3. The activities that require detailed tutorials or mini workshops
4. The teaching interventions and writing assignments necessary to guide student learning
5. The role of reflective activities in the development of student deliverables
6. How the learning activities will foster an inclusive experience

Education starts here:

- What are the knowledge/skill bottlenecks students will encounter during the CURE? What tasks or expectations might represent obstacles?
- What does group work contribute to this activity? Which activities require group work?
  - Is the amount of work simply too much for a single person?
  - Is the interaction among students necessary to generate answers?
- How does students’ ability to work well with others factor into the goals of the CURE?
- What does an authentic research team structure resemble in the field of research associated with the CURE? How can the CURE mimic this?
- How will you design groups whose structure promotes inclusivity and constructive intellectual exchanges?
- Are there activities that require tutorial or workshops to be explained?
  - Can students perform all necessary analyses for the CURE without particular training?
  - Will students need to be introduced to a particular software program to collect or analyze data? To construct figures or tables?
  - Can you feasibly guide individual students/groups of students through the experimental components of the CURE without prior training on a smaller/published/example dataset?
- How are you introducing the expectations for graded assignments and deliverables to students? How do they know what is expected of them?
- What are the necessary or helpful intermediate steps to completing all deliverables and graded assignments? What steps do you go through to complete these activities yourself? What about a graduate student? How would you guide a mentored undergraduate researchers outside of the course through this activity? Based on all of this information, ask yourself: “What would a novice

do?"
- How can reflection help students develop better deliverables? What is the role of reflection in the development of student deliverables?
- Can a think-pair-share structure facilitate progress or foster the development of deliverables?
- How can you design learning activities that foster an inclusive experience?

Scholarship starts here:

- What are the different steps of the research process that students will be going through? What numbered elements of **Figure 2** are necessary or, instead, need to be edited?
- What foundational skills and knowledge of the field of research should be incorporated in the CURE? Does that include skills or knowledge that may not in fact be strictly necessary for the specific research question/hypotheses tested in the CURE?
- Which of the CURE tasks is typically achieved by individuals in a research setting? By groups of people?
- Are certain elements of reflection or of the scaffolding of the research project typically shared through publication (for example as appendices) in professional deliverables?

Research and Education together:

- Instructors often do not consciously realize that the process of research relies upon underlying concepts and practices. Researchers familiar with the research process skip over some of these foundations in their daily engagement in scholarship. Instructors can help students become researchers by explaining the critical concepts underlying the practice of research ([**Activity3**]).
- Engaging in research can be daunting for students. Reflection workbooks and journaling can help students overcome the challenge of the research process, reduce their anxiety, increase their appreciation of research, and facilitate exchanges between students and instructors (Apgar, 2022). Research shows that training significantly improves students' reflection skills (Devi et al., 2017).
- The entire course would benefit from a scaffold that guides students (and instructors) through the learning process. Sabel (2020) provides an example of a framework to develop the scaffold of a course. Another example is showed by the Decoding the Disciplines framework (https://decodingthedisciplines.org/) introduced here. [**Appendix 1**] shows yet another example in which students, for each element of the CURE (e.g., searching the literature, data collection, data analysis, writing the material and methods) engage in a series of activities that lead them to produce a research deliverable (**Fig. 2**).
- Group meeting agendas can help students prepare the meetings they will have with their teammates during class time. Example of questions and guidelines to prepare various group meetings are provided in **[Activity 14]**.

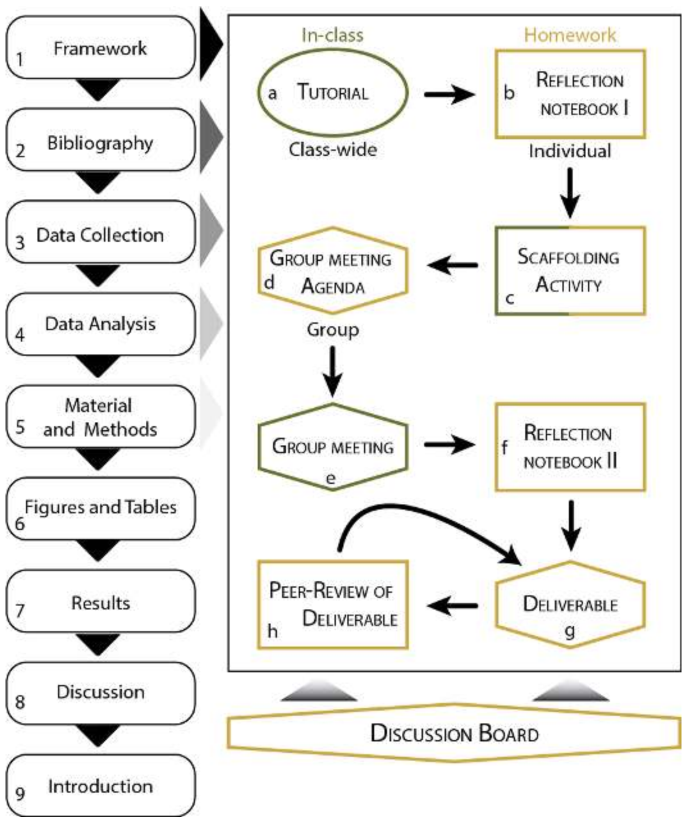

Figure 2. Possible CURE scaffold. The colors denote the location of the work, the shape of the boxes whether the work is undertaken individually, as part of the research group, or class wide. Arrows show the flow of work through deliverables for which prompts are provided in [Activities 15-17].

- There is an extensive literature on the design of groups emphasizing the need to carefully consider the group composition as well as the framing of the group work by the instructor(s). Best practices for group work are summarized in Wilson et al. (2018) and at https://lse.ascb.org/evidence-based-teaching-guides/group-work/
  - Consider carefully the size of the groups (Heller & Hollabaugh, 1992).
  - Consider carefully the gender and ethnic minority status of students in the composition of the groups (Adams et al., 2002; Micari & Drane, 2011).
  - Assigning specific roles to students is helpful to encouraging and structuring the conversation (Heller & Hollabaugh, 1992). Enforcing a rotation between the roles of the students, if possible, can be helpful. There are several strategies for distributing roles presented in the literature (e.g., Olimpo et al., 2016; Winkelmann et al., 2015).
  - Including discussions of the nature of intelligence, academic failure, systemic biases, as well as fostering a growth-mindset in students can greatly help students overcome social-comparison concerns (Micari & Pazos, 2014).
  - Setting group goals can also be helpful (Werth et al., 2022) and strategies for effective group work are important (Washington University in St. Louis). In addition to assigning specific roles, instructors should consider using group contracts [**Activity 18**], and peer evaluations (see [**Activity 19**], which have been showed to encourage student participation (Chang & Brickman, 2018), can facilitate the assessment of the contract compliance, and make easier the identification of conflicts and problems leading to a more rapid resolution. Group contracts and peer evaluations can be used as part of the grading scheme to determine individual contributions.
  - The Peer Assessment Factor is a quantitative assessment of the students by themselves and peers that can be used as a formative assessment and a guide for interventions.
  - Examples of other activities, and interventions are provided in the PETS Process Manual.
- Frameworks for mini-workshops are presented in the primary literature for many tools and protocols. Some are specifically aimed at CUREs. A statistical workshop is presented by Olimpo et al. (2018). Sewall et al. (2020) presents several mini-workshops on computational tools (R and QIIME2). Alternatively, published protocols might be sufficient to guide students through an experiment or research task without the need for a prior tutorial or workshop (Acuna et al., 2020; Buser et al., 2020; Chen, 2018; Craig, 2017; Goeltz & Cuevas, 2021). Several published CUREs include within the supplementary information files useful protocols and tutorials (e.g., Bucklin & Mauger, 2022; Jurgensen et al., 2021; Poole et al., 2022; Roberts et al., 2019; Zelaya et al., 2022).

- There are numerous published strategies to help scaffold inquiry activities such as authentic research. Several are summarized in a TLRC teaching topic. Additional information on scaffolding is provided in Killpack et al. (2020).
- Writing-to-Learn (WTL) activities can help students build their understanding of the existing literature on the CURE topic, work through their project, and complete their deliverables (Balgopal et al., 2018; Bangert-Drowns et al., 2004; Fry & Villagomez, 2012). Many examples of WTL activities are available online (see in particular the Center for the Study and Teaching of Writinghttps://cstw.osu.edu/writing-learn-critical-thinking-activities-any-classroom). A selection of scaffolding activities derived from WTL concepts that can be implemented in a CURE are presented here, including an activity guiding students through the process of deciding on their data analysis protocol [**Activity 20]**, an activity helping students summarize the existing literature [**Activity 21**], two sister activities helping students compare their findings to data from the primary literature [**Activities 22-23**], and an activity meant to help students prepare the discussion section of their paper **[Activity 24]**.
- Tutorials for writing can be developed for each section of the deliverable, including the elements of an IMRaD paper (Introduction, Methods, Results, and Discussion). An example of the structure of such tutorials are showed in **[Activity 16]**. An example activity to develop students' graphical skills is showed in **[Activity 25]**.
- Other activities can be designed to scaffold the students' work including activities encouraging students to predict their results and represent them graphically or activities comparing the introduction and discussion from a single publication to emphasize their roles as bookends of the deliverable.
- **Figure 2** shows a possible framework by which reflection can be incorporated in the scaffolding of the course to enable students to build their metacognition while they engage in scholarship (see Denke et al., 2020). Student reflection can be guided by a set of questions and an information literacy framework. An example of such guide is provided in **[Activity 26]**.
- Best practices to build student metacognition are reviewed in Stanton et al. (2021) and at https://lse.ascb.org/evidence-based-teaching-guides/student-metacognition/.
- Think-Pair-Share activities, whereby students first reflect upon their work and then discuss their thoughts with group members before to engage with the rest of the class and the instructor(s), may be helpful in promoting reflection as well as conversations among students; they can also be used as formative assessments by the instructor(s) to improve student performance (Akhtar & Saeed, 2020). Best practices for this active-learning activity have recently been reviewed in Cooper et al. (2021) and Prahl (2017).
- The TILT framework (Winkelmes et al., 2016) enables instructors to increase transparency in the assignments by providing clear descriptions of the purpose, tasks, and evaluation criteria for each activity the students engage in. Examples of applications of this framework are provided at: https://tilthighered.com/tiltexamplesandresources.
- There are numerous sets of best practices and recommendations (e.g., Cooper et al., 2020b; Faulkner et al., 2021; Linder et al., 2015; see question 10 on p. 104) that have been developed to promote an inclusive classroom environment (whether in a CURE or not). Miller et al. (2022) presented a framework to incorporate "antiracist, just, equitable, diverse, and inclusive principles" in the design of a CURE. Best practices for inclusive teaching are presented in Dewsbury and Brame (2019) and at https://lse.ascb.org/evidence-based-teaching-guides/inclusive-teaching/.

Box 5: Important points

- The prior knowledge of the students informs the first rungs of the learning scaffold.
- The ELOs for the course provide the framework for the endpoint of the scaffold.
- Group work should be intentionally designed, supervised, and assessed to lead to accountability and personal learning milestones.
- The scaffold adopted should provide students all required background information, guide them through challenging tasks in an accessible context, provide opportunities for reflection, and enable students to repeat or expand on their work to advance the research process.
- Writing-to-learn activities can facilitate student learning and development as writers.
- Information literacy is critical to student success and can be fostered through reflective work as well as activities promoting metacognition.
- The Transparency framework of TILT enables instructors to articulate the purpose, tasks, and criteria for evaluations of each activity to students, thereby promoting metacognition and reflection.

6. How will students communicate the results of their research?

Overview: The communication of research findings is critical to the process of scholarship, including undergraduate-led research (Spronken-Smith et al., 2013). It is also an important element of CUREs. Many traditional outlets of professional scholars are also appropriate for student research derived from CURE. Thus, presenting research findings at conferences has been showed to greatly benefit undergraduate students (Little, 2020). Publishing the findings of a CURE on a database used by researchers has been demonstrated to lead to increased student motivation (Wiley & Stover, 2014). Finally, publications resulting from CUREs have been associated with both personal and professional benefits for students (Turner et al., 2021). In all forms of deliverables, it is important to consider how the contributions of all members of the CURE will be recognized. Research has showed that there exists a gap between what an instructor or professional researcher might consider necessary for authorship *versus* what students consider necessary (Turner et al., 2021). This section enables you to determine:

1. The appropriate mode(s) of dissemination of the CURE's findings
2. The division of labor among students in preparing the deliverable
3. Authorship rules

Education starts here:

- What is an appropriate expectation in terms of deliverable given the time constraints of the CURE?
- Would a specific deliverable be more motivating for students? Have you asked students?
- Is one format of deliverable going to benefit students professionally more than another?
- Which mode of dissemination would be appropriate for a professional scholar at this point in the research? Would that be any different for a student participating in a mentored research experience?
- Will the deliverable be evaluated for a grade?
- Will each student be expected to produce a final document (be it a dataset, poster, manuscript, or proposal) or will the deliverable be prepared by a group of students?
  - Will each student prepare one section of the deliverable? Alternatively, will all students be contributing to all sections of the document?
  - Will all students in the group be evaluated as a group with a single grade for the document applying to all students? Alternatively, will the students receive an individualized grade?
  - If students will be individually graded, what criteria will be retained to differentiate between group members? Are those going to impact authorship order?
- How will the intellectual and practical contributions of the students participating in the CURE be recognized in products of the project?

Scholarship starts here:

- What deliverable are you expecting/hoping for at this stage of the research?
  - Are the students completing a project leading to a publication manuscript? Will the students produce a version of the manuscript near-ready for submission or will there be large amounts of work undertaken after the conclusion of the course to bring the manuscript to a submittable form? Who will undertake this work?
  - Are the students helping start a new project, developing preliminary results for a proposal or contribute to a database?
- Will the findings of the CURE be appropriate for a conference presentation?
- What are the standards for manuscript authorship in your field? At your institution? In your research group? Alternatively, when are contributors merely acknowledged?
- How will the contributions of instructors and outside researchers be considered for authorship?

Research and Education together:

- Different deliverable formats are not mutually exclusive.
- The format of the CURE deliverable should reflect the authentic mode of delivery of research findings in the field (Kloser et al., 2011).
- Students enjoy the opportunity to present the results of their research at conferences and/or publish their work (Amir et al. 2022).
- Participation in research conferences has been showed to benefit students both by helping students develop their communication and research skills, but also by positively impacting their career and engagement in other activities (Little, 2020).
- The opportunity to publish the findings of the CURE enhances student motivation (Wiley & Stover, 2014).
- Publications can be powerful deliverable for a CURE because:
  - They can act as scaffold with the different sections of a manuscript corresponding to elements of the research process
  - They represent the scholarly endeavor and can help students gain an appreciation for the process of research.
  - They can help build the students' identity as researchers through co-authorship of a published document.
  - They positively impact the careers of students (Turner et al., 2021).

- They can be motivating and thus increase student engagement in the CURE.
- They provide a tool for accountability among students and with the instructor.
- Consider how the participation in a national or university-wide CURE program may dictate the deliverable if the CURE is not researcher-driven.
- Undergraduate research symposia can be great venues for the students to present the results of the CURE. Many colleges and universities as well as some departments at larger institutions organize symposia at least once a year (Presentation Opportunities).
- There are also numerous national conferences dedicated to student research (e.g., SigmaXi; Council on Undergraduate Research).
- In addition to discipline-specific publication venues and journals aimed at professional scholars, there are also open-access online undergraduate research journals (Sun et al., 2020) that may be appropriate outlets for the CURE's findings.
- The Knowledge Bank of Ohio State (https://library.osu.edu/kb) enables the archiving of intellectual outputs of the university community in an accessible digital format. Other digital repository, some discipline-specific, may also be appropriate (e.g., the Environmental Data Initiative)
- There are online repositories of research products that are citable and can be appropriate for the results of CUREs (e.g., https://figshare.com/).
- Digital and artefactual exhibits can also appropriate venues for the presentation of archival and object or specimen-based research (e.g., Donegan et al., 2022; ENG5612 at Ohio State).
- Multiple online scholarly projects and databases (**Table 13**) may be appropriate venues for student contributions (e.g., Map of Early Modern London).
- Establishing clear authorship guidelines is critical to a smooth and rewarding experience:
  - Students should buy into the rules. It is important to explain the rationale for authorship rules to students.
  - Students may also have valuable input on the rules. Consider developing authorship rules specific to the CURE, with the students, that incorporate guidelines and recommendations from relevant institutions as well as best practices in the CURE's field of research.
- Many universities and colleges have developed authorship guidelines including The Ohio State University (Authorship Guidelines) and other institutions (e.g., Harvard University, University of Michigan, Yale University).
- Many professional societies have also developed best practices documents relating to authorship (e.g., American Physical Society, American Psychological Association, British Sociological Association).
- Some journals and publishers also have rules for authorship (e.g., Nature, British Medical Journal, Taylor & Francis).
- If other instructors or researchers (e.g., graduate students, peer teaching assistants, research collaborators at other universities) will be involved in the project, they should be part of the conversation and agreement on the authorship rules. Consider the authorship rules you employ in your research group.
- There are several published sets of guidelines for authorship that can also be used. The CRediT taxonomy (Honoré et al., 2020), which has been adopted by PLOS journals (PLOS Authorship Guidelines), can provide a starting point to develop guidelines specific to your CURE.
- The copyright services department of the Libraries (https://library.osu.edu/copyright) can help with issues of students' intellectual property rights.
- In the case of CURE-specific requirements for authorship, consider whether or not it is appropriate to tie authorship to course requirements and achievements (e.g., fulfilling the contract if contract grading, obtaining a certain minimum grade on the deliverable).
- Students should be offered the opportunity to approve the final version of the manuscript they co-author, even after completion of the course.

Box 6: Important points

- Different deliverable formats are not necessarily mutually exclusive.
- Establishing clear guidelines and requirements for authorship is critical.
- Communicating the results of the CURE should reflect authentic scholarly communication in the field.
- Publication manuscripts can serve as scaffold, provide motivation, ensure accountability, and shape student identity.

7. How will the progress and experience of students be assessed?

Overview: The assessment of the students' work in a CURE enables the instructor and the entire class to stay up to date on the research progress, including the findings of the project and the setbacks. Well-designed assessments also enable students to reflect on their learning, help them identify their difficulties, and provides tools for overcoming obstacles. Because CUREs are elements of formal courses, it is also often necessary to evaluate the students' work qualitatively and quantitatively for the purpose of a grade. This section enables you to determine:

1. The goals of assessing the CURE work
2. The mode of grading to adopt in the CURE
3. The proportion of the evaluation represented by group work and individual work
4. The proportions of formative assessments and summative assessments
5. The roles of instructors, peers, and self in evaluation.
6. The nature of the assignments graded
7. The need for specific rubrics and grading criteria
8. The importance of an inclusive mode of grading

Education starts here:

- Does all evaluation work have to translate to a grade?
  - Can some assignments be evaluated simply for completion/genuine participation?
  - Can some scaffolding assignments be reviewed to provide feedbacks without associating the work with a grade?
- What are the goals of the grading in the CURE?
  - Will it be used to determine authorship eligibility?
  - How will the grading be aligned with the expected learning outcomes (ELOs) for the CURE?
- Which assignments will be graded?
  - When you assess student performance, what are you rewarding? Is the grading based on the mastery of the ELOs or on other criteria? Are those criteria made explicit to students?
  - Are students given a chance to revise their work and correct their mistakes?
  - How do you give students practice with and feedback on performance items that you are rewarding on the final assignment?
  - Do you give students the opportunity to reflect on what they are learning or how they are growing?
- What percentage of the class grade will be represented by the CURE grade?
  - How does this compare to the time investment made by students?
- Will group assignments be graded or will only individual assignments be graded?
  - When group assignments are graded, do all members of the group get the same grade?
  - Does the mode of grading lead to increased intra-group conflicts?
- What mode of grading will be adopted in the CURE?
- What proportion of the final grade for the CURE is represented by formative assessments *versus* summative assessments?
- Are summative assessments based on the formative assessments on which the students got feedback?
- Is there room for self-evaluation in the CURE? What about peer-evaluation beyond peer-reviews of deliverables?
- Can students grade some of their own assignments with the help of a rubric as a metacognition tool? Can this exercise be combined with instructor feedback and a redo enabling students to gain missed points?
- How often will students be completing work reviewed by the instructor(s)?
  - How often do students get instructor feedback?
  - How many iterations of a given document, section of deliverable, research product will students be producing?

- Will students get rubrics for each assignment?
  - How will explicit grading criteria be provided?
  - Is it possible to hand out examples of successful products to students as models and examples of works that do not meet expectations to guide their own writing?
- What approaches can you adopt to ensure an equitable and inclusive mode of grading? How can assessments be used to reduce the equity gap?

Scholarship starts here:

- How are scholars evaluated in your field? Can this mode of evaluation be mimicked in the CURE?
- What determines the quality of a deliverable in your field and what aspects of that document quality do students have control over?
- What is a reasonable quality standard to expect for the deliverable of choice? Is it realistic to expect the students to produce a document nearly ready for publication submission or presentation?
- How would you assess the work of a mentored research student engaged in a similar project? What expectation would you have of this student's deliverable?
- How will the success of the research impact assessment and grading?

Research and Education together:

- The nature of CUREs as authentic research may lead to failures and mistakes; in fact the novel aspect of the CURE will almost certainly lead down unpredictable paths and outcomes. Designing the assessment of the CURE to mitigate the effects of technical problems, negative results, and the steep learning curves of scholarly endeavors is critical to student engagement. The goal of assessments should be to inform students and instructors alike of the progress of students as researchers, not the success of the observations or experiments. The ability to troubleshoot, the resilience of students in the face of failures, the understanding and explanation of errors, and the repetition of tasks should be integrated in the grading scheme. Finding the solution to a problem and understanding why an experiment did not work are progress. A discussion of failure, how to approach failure with students, and the point of view of the instructor in the context of CUREs is provided in Townsend (2022).
- Backward-design requires the alignment of the assessments with the ELOs for the CURE. Instructors should rigorously verify that all ELOs for the CURE are assessed by the end of the course, but also that assessments do not evaluate expectations that are not made explicit to students. Different assessment modes are suitable for diverse learning goals (Verb wheel).
- Transparency in assessment is critical to student success and inclusive teaching. The TILT framework (Winkelmes et al., 2016) presented earlier in this document is particularly helpful: https://tilthighered.com/tiltexamplesandresources.
- Beyond traditional point-based grading, there are other modes of grading that an instructor can consider.
  - Criterion-referenced grading provides students with flexibility in the weights of the various components of the course. It enables students to mitigate test anxiety and emphasize formative assessments; it can also be used by students to lessen the consequences of out-of-school responsibilities on their academic endeavors. In practice, you should determine bounds for the weights of the different categories of course assessments and enable students to develop their own formula for the CURE grade. Every student will be graded according to their unique formula (which can easily be setup in a spreadsheet program).
  - Contract grading (Inoue, 2019) offers the opportunity to combine flexibility for students and for instructors while maintaining rigor in expectations and avoiding conflicts over grades. Contract grading is a format of grading that does not involve points or letter grades, apart from the final course grade. At the start of the semester, students choose/agree to/sign a contract that sets their path for the semester. Each contract lists the work required of the students for a particular outcome (e.g., an A, a B). If the student satisfactorily completes the work associated with their contract, they will earn that grade. Contracts are often associated with a mid-semester conference to check on progress towards the completion of the contract and reevaluate commitments if necessary. This conference is repeated at the end of the semester to assign the students' grades. The onus of tracking contract compliance may be placed on the students who are responsible for writing self-evaluations ahead of each conference in which they need to address their work, its quality, their engagement with the class, etc. You should respond to these self-assessments during the conference. You can always disagree with the student in their final assessment of their performance. Students can also be asked to keep track of their time using a labor log. An example of contract grading for writing courses is provided by Inoue (2019). Another example been published online by Cathy N. Davidson. Contracts can be more or less complex incorporating aspects of criterion-referenced grading (Hiller and Hietapelto 2001) or including specifications grading elements (Lindemann and Harbke 2011; see **Appendix 2** for a model applied to CUREs).
  - Specifications grading can help uphold academic standards and motivate students while reducing grade anxiety and cheating. In specifications grading, the assignments are graded on a pass/fail basis, a check/check minus/check plus/unsatisfactory basis, or an excellent/meets expectations/needs revisions/fragmentary basis. Instructors should provide clear specifications of what constitutes an acceptable work, what qualifies for a check (or earns a check plus), what leads one to meet (or exceed) expectations, etc. Students may also be given the opportunity to revise their work. Assignments may be assessed individually or grouped into modules that are assessed holistically. Modules can be weighed to reflect the complexity of the tasks, their relevance to learning outcomes, and/or their status as formative/summative assessments. Several models of specification grading for different fields have been published (e.g., Blackstone & Oldmixon, 2019; Carlisle, 2020; Elkins, 2016; Howitz et al., 2021).
- Group reviews can be used to assess the contributions of all members of a group to the outputs of the CURE, including specific documents. A group review template is provided in [**Activity 19**]; others exist in the literature (e.g., Bucklin & Mauger, 2022; Waddell et al., 2021).
- Self-evaluation, also called self-assessment or self-grading can be a powerful tool to promote student pacing and success, primarily as part of formative assessment (Andrade, 2019).
- Many rubrics have been published online (University of Minnesota, Chicago State University) and in the peer-reviewed literature for both writing assignments and CUREs (Bakshi et al., 2016; Kishbaugh et al., 2012; Lee & Le, 2018; Murren et al., 2019; Ramírez-Lugo et al., 2021; Sewall et al., 2020; Waddell et al., 2021).
  - An example of a rubric for discussion boards in provided in **Appendix 3.**
  - An example rubric for a manuscript-type deliverable is provided as part of the peer-review presented in **[Activity 12]**.
  - Bucklin & Mauger (2022) as well as Merrell et al. (2022) both include rubrics for a poster presentation stemming from CUREs.
  - An example rubric for a grant proposal-type deliverable is provided in Rennhack et al. (2020).
- When designing a rubric, consider the following best practices (from Bean, 2011):
  - Numbers on the rubric do not add up to 100 and do not represent directly course points.
  - The grade associated with the rubric is always presented as a letter or a check/check plus/check minus, not a number.
  - Students are explained how the rubric is used in grading.
- A good rubric should include a detailed explanation of the task students are expected to perform, the characteristics of the work that will be evaluated, the levels of mastery that will be considered, and descriptions of the characteristics for each level of expertise. Additional guidelines for the development of rubrics can be found in a number of different publications (Boston University, Brown University, and Arizona State University). You can also consider co-creating the rubric you will use with the students (University of Colorado Boulder).
- Highly structured courses with frequent low stakes assignments increase student engagement (Cavinato et al., 2021), lead to higher performance (Freeman et al., 2011), and help reduce the equity gap (Eddy & Hogan, 2014; Haak et al., 2011).
- Consider using concept inventories to test for knowledge acquisition in the field of research (**Table 6**).
- (More) authentic formative assessments can be undertaken by reviewing and grading lab notebooks, periodic research updates akin to those a research student would provide in a lab/research group meeting, conference-style abstracts, elevator speeches, or chalk board presentations.

Box 7: Important points

- Assessments should be aligned with ELOs for the CURE.
- Rubrics and clear criteria for success help communicate the expected level of mastery to students.
- Student practice and instructor feedback on formative assessments should be aligned with summative assessments.
- Contract grading, specifications grading, peer-grading, and self-assessment are valid alternatives to traditional numerical grading by instructor.
- Frequent low-stakes assignments can help reduce the equity gap.
- Accountability for personal tasks and peer/self-review facilitate grading of group work.

8. How will research learning tasks change as discoveries are made and initial research questions are answered?

Overview: The process of research is inherently iterative and involves sequential hypothesis testing. As results emerge from observations, experiments, and analyses, new hypotheses are developed and require testing. This is often the case in CUREs. Unlike a scholar's research program, however, CUREs have strict curricular goals and involve time limits, including a comparatively small number of weekly hours of research engagement and an end to the research process imposed by the end of the instructional period be it a quarter, semester, or other. This section enables you to determine:

1. Whether to include such sequential aspect of research within a single implementation of a CURE or across repetitions of a CURE.
2. How to bring about the transition from one hypothesis to another
3. The need to revise the scaffold and learning tasks of the CURE to fit new hypotheses and research paradigms
4. The importance of planning the research course of the CURE to offer an original experience to successive cohorts of students

Education starts here:

- Will students have time to explore only one hypothesis/question, or will they be able to at least partially engage in a second one?
- For students exploring researcher-chosen questions/hypotheses:
  - Will they be able to suggest their own follow up hypothesis/question?
  - How will you explain to them how the question/hypothesis they are investigating was developed?
- When revising or repeating a CURE:
  - What elements of the CURE are outdated? Any recommended paper or analysis no longer reflecting the knowledge or best practices in the field?
  - How will you ensure that the upcoming CURE is not merely a variation on a previous iteration, in which all novelty has been lost?
  - What should you change to make sure that the CURE does not become a “cookbook CURE”, a research experience for which the approach and protocols are fully developed and the result guaranteed, leading students to engage in research novel only to them that does not significantly contribute to the field?
  - Is the new iteration of the CURE following-up on questions or hypotheses developed in previous versions of the course?
  - What advances in the field have been made since the last time the CURE was run? Should planned questions/hypotheses be revised as a consequence?
- Have new best practices in pedagogy and CURE implementation been published since the last iteration of the CURE? Do they mandate revisions to the structure, scaffold, or assessment of the CURE itself?

Scholarship starts here:

- Does the scope of the CURE represent a publishable manuscript in the field of research of the CURE? Will more than one iteration of the CURE be necessary to answer all necessary questions/test all necessary hypotheses?
- Are adjacent or related questions/hypotheses being tested by colleagues, collaborators, research students, thus limiting the scope of the work the students can follow-up on?
- When developing a new researcher-driven CURE:
  - What is the place of this CURE within the research program of the researcher(s) involved?
  - What are the natural follow-up hypotheses or questions already known?

Research and Education together:

- It is critical to introduce students to the nature of the scholarly endeavor including the fact that research generates more questions than answers and the need for researchers to critically select the questions that they will in fact explore. This discussion can take place as part of the process to select the question investigated in the CURE or not (see [**Activity 2**]). The roles of funding agencies, other scholars, institutions, and systemic biases in this selection should also be presented. Some aspects of these issues are discussed in https://opentextbc.ca/researchmethods.
- It is possible to demonstrate to students the nature of research and the succession of investigations that lead scholars to build piece-by-piece the puzzle of a particular topic without requiring students to walk these steps themselves through retrospectives and walkthroughs of the history of the current paradigms and questions they will explore.
- CUREs can build on each other over time to enable the investigation of particular topic through a series of studies (Satusky et al., 2022; Sun et al., 2020: figure 2). It is important to incorporate this history when presenting the CURE to students.
- Extensive notetaking during the design and implementation of the CURE enables an instructor to revise the CURE. There are published guides to revising courses and reflecting on pedagogy (see for example McGahan, 2018).
- Instructors teaching researcher-independent CUREs should consider following the research advances in the field of the CURE as they do their scholarship area to maintain up-to-date knowledge of the field, active areas of research, literature reviews, and methodological developments.

Box 8: Important points

- Just like a research program, a CURE changes over time following research inquiry, progress, and setbacks.
- Showing/explaining the development of the CURE overtime to students is integral to showing/explaining the research process.
- Just like the work of students during the CURE is iterative, so is the work of (re)designing the CURE itself.
- Beware of creeping away from a CURE towards a “cookbook lab” with successive iterations.

9. What are the logistical obstacles and solutions for the different steps of the CURE?

Overview: The implementation of a CURE requires students to access the tools of the research trade. These are discipline-dependent, but may include lab space, consumables, specific technologies or equipment, library and documentary resources, research specimens, and computing facilities. Certain CUREs may also involve field experiences, which introduce their own set of logistical challenges. Planning the needs of the CURE is critical to its success and may influence the nature of the research questions investigated with students. This section enables you to determine:

1. The resources necessary for data collection
2. Alternative sources of data
3. The needs of the data analysis
4. The potential for crowd-sourcing the CURE’s support
5. Possible sources of funding to support CUREs

Education starts here:

- What is the budget of the course?
- Is the CURE part of the broader impacts of a grant?
- Is the CURE part of a creative teaching or curriculum redesign effort that can receive funding or logistical support from the institution, professional societies, or funding agencies?
- What help can support staff, including lab technicians, research and teaching assistants, lab coordinators, librarians, IT staff, and museum staff, provide with data collection and analysis? What do they need to be able to help?
- Who can help me think about my CURE development process and the pedagogy of this model of research?

Scholarship starts here:

- What are the consumable needs of the CURE?
- Do loans of specimens need to be secured prior to the start of the CURE? What are the restrictions on the handling of research specimens and materials?
- What equipment, including laboratory equipment and computational resources inclusive of hardware and software programs, is necessary to not only collect and analyze the data, but also prepare deliverables? Is this equipment accessible in teaching or research facilities on campus?
- What permits, certifications, and approvals are necessary for students and instructor(s) to undertake the research?
- What existing databases and online datasets of images, observations, measurements, etc. can be leveraged to facilitate the data collection for the CURE?
- For researcher-driven CUREs:
  - Can the CURE be integrated with other data collection efforts of the research group? What are the benefits for the research students involved in contributing their data to the CURE?
  - What is the role of research collaborators in the project?
- Are there sources of funding associated with the research ongoing in the laboratory/research group that can legitimately support the research undertaken in the CURE?

Research and Education together:

- Although the cost of a CURE has been reported to be higher than that of a traditional introductory science laboratory (Rodenbusch et al., 2016; Spell et al., 2014), it is much lower than the cost of mentored research experiences or summer research experiences (Rodenbusch et al. 2016; Smith et al., 2021). Some estimates are around $400 to $500 per student per course and many CUREs are cheaper (Poole et al., 2022) or even approach no financial costs.
- Online databases (**Table 13**) and freeware programs (**Table 14**) enable data collection and analyses at low costs.
- Collaborators and colleagues may be able to share equipment and supplies at low costs. CUREs can be departmental/institutional resources that spur enrollments and raise the profile of the teaching/research unit. Discuss with stakeholders (e.g., department chair, associate dean, and colleagues) the benefits of supporting the CURE.
- Many institutions have some equipment and resources (e.g., supercomputer, computer labs, greenhouse space, imaging facility) whose costs and access are mutualized or free for in-house projects.
- The integration of research and teaching missions of the CURE may enable different sources of funding to support the work (e.g., CUREnet). Those may include grants from professional societies, government agencies, internal competitions at the host institution, etc. (e.g., Council on Undergraduate Research). There are also calls for proposals appropriate for CUREs (NSF Division of Undergraduate Education). Some network CUREs may provide seed funding to implement the course (DeChenne-Peters & Scheuermann, 2022).
- Trainings, permits, and approvals should be secured ahead of the CURE as much as possible. Student-specific trainings should be integrated in the course. Logistical obstacles of research are important part of the curriculum. Students learn of the realities of the process of research and the important legal and ethical regulations that are associated with it. Additionally, students who navigate through these obstacles may gain important skills in project management.
- The involvement of research group members in the CURE should be designed to benefit them in the form of co-authorship, mentoring experience, opportunities for career advancement, funding, etc.
- Teaching assistants, including graduate and undergraduate students, have been showed to be very helpful in supporting undergraduate students enrolled in CUREs (Olson et al., 2022). They should receive necessary training to learn how to teach a CURE (Kern & Olimpo, 2022)
- There are multiple offices and resources at the Ohio State University to support the development of CUREs that are summarized in **Table 15**.

Workshops and Endorsements	
Teaching Information Literacy	https://drakeinstitute.osu.edu/endorsement/teaching-information-literacy https://u.osu.edu/teachinginfolit/teaching-information-literacy-workshops/on-demand-asynchronous-workshops/
Meaningful Inquiry	https://drakeinstitute.osu.edu/endorsement/meaningful-inquiry
Teaching through Writing	https://drakeinstitute.osu.edu/endorsement/teaching-through-writing
Inclusive Teaching	https://drakeinstitute.osu.edu/endorsement/inclusive-teaching
Research Mentoring	https://drakeinstitute.osu.edu/endorsement/research-mentoring-training
Digital Humanities Pedagogy	https://drakeinstitute.osu.edu/endorsement/digital-humanities-pedagogy
Course Design Institute	
Drake Institute Course Design Institute	https://drakeinstitute.osu.edu/instructional-support/course-design-institute
Pedagogical resources	
The Teaching and Learning Resource Center	https://teaching.resources.osu.edu/
Instructor Resources at University Libraries	https://library.osu.edu/instructor-resources-at-university-libraries
Offices and Professionals	
Office of Undergraduate Research and Creative Inquiry	https://ugresearch.osu.edu/
Michael V. Drake Institute for Teaching and Learning	https://drakeinstitute.osu.edu/
Office of Academic Enrichment	https://oaa.osu.edu/oae/
Honors and Scholars Center	https://honors-scholars.osu.edu/
University Libraries	https://library.osu.edu/instructor-resources-at-university-libraries
Office of Service-Learning	https://service-learning.osu.edu/

Table 15. Resources at the Ohio State University for the development of CUREs.

Box 9: Important points

- Obstacles to data collection can be overcome with research databases, citizen science data, museum databases, and online sets of images and observations or measurements.
- Campus resources including IT, the Libraries system, as well as research lab members and collaborators can help overcome data collection and analysis obstacles.
- CUREs can be integrated with mentored student research (of graduate or undergraduate students) to facilitate funding, mentoring, data collection, data analysis, and the professional development of research students.

10. What are the roles of instructional and support staff?

Overview: Although it may be tempting to think of the instructor (or instructors) as the linchpin of both the research and pedagogical processes of the CURE, the structure of CUREs is inherently student focused. As such, CUREs provide the opportunity for instructors to rethink their identity in the classroom to include a critical mentorship component built around providing a supportive environment in which students are assisted in acquiring knowledge and skills, but also learn the significance of their work (**Figure 3**).

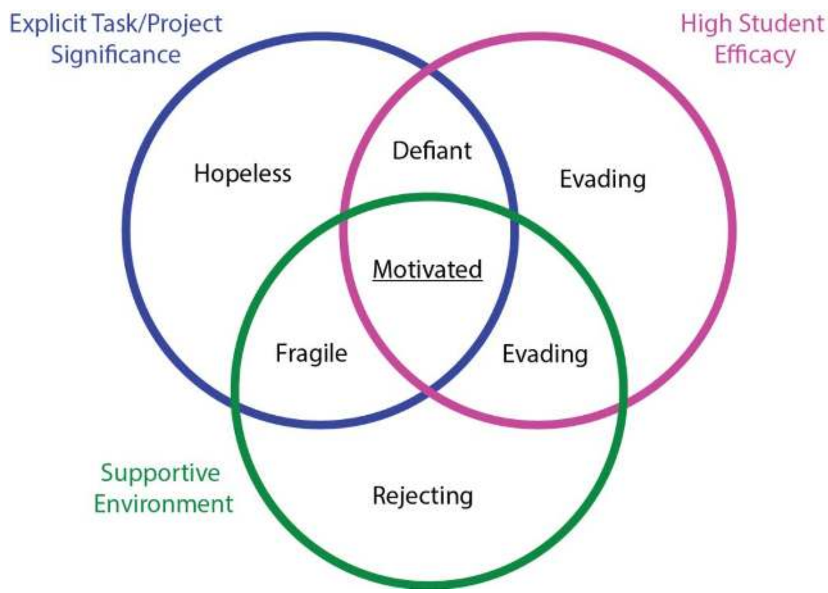

Figure 3. The three components of student attitudes that should be fostered by research mentors (redrawn and modified from Ambrose et al., 2010).

A successful CURE is easier to implement with the support of other members of the research team, educational support staff, or campus community. A researcher-independent CURE may rely on the resources and structure of a national program, but the success of the CURE will also likely involve campus members who can complete and enhance the work of the instructor. This section enables you to determine:

1. The role(s) of the instructor(s) in mentoring the CURE
2. The fostering of a supportive, motivating environment
3. The role of the instructor and other experienced scholars in the research
4. The expectation and needs of graduate or peer-teaching assistants
5. The support provided by lab managers, technicians, and other staff members

Education starts here:

- How can the experience of CURE students be improved through interactions with students who have previously completed the course, mentored undergraduate researchers, graduate students, or postdoctoral researchers?
- What challenges are students likely to encounter during the CURE?
- What are the obstacles to the mentor-mentee relationship you foresee for the CURE? Consider expert-novice gap, communication skills, and cultural differences as well as “social-distance” (Shanahan, 2018).
- What is the ability and willingness of instructors to commit time to mentoring?
- What is the prior experience of the instructor(s) in mentoring research?
- What training and support is necessary to foster the success of postdoctoral researchers, graduate student instructors, or peer undergraduate assistants as educators?
- How do the ELOs of the CURE impact the model of mentoring that instructors should approach?
- What are the expectations for postdoctoral researchers, graduate student instructors, or peer undergraduate assistants of the instructor-of-record? Of the department/college/graduate school/HR/etc.?
- What are the ELOs as well as professional and personal goals of postdoctoral researchers, graduate student instructors, or peer undergraduate assistants?
- How can the mentoring of postdoctoral researchers, graduate student instructors, or peer undergraduate assistants by experienced researchers and teachers help them reach their goals?
- What are the roles of the support staff and outside researchers in the CURE? Can these professionals free up instructor time for mentoring by assuming some technical responsibilities?

Scholarship starts here:

- What is the commitment (time and effort) that the instructor will dedicate to the research, particularly in class, that cannot be directed to mentoring?
- What is the responsibility of postdoctoral researchers, graduate student instructors, or peer undergraduate assistants in the research tasks of the CURE?
- What training and support is necessary to foster the success of postdoctoral researchers, graduate student instructors, or peer undergraduate assistants as researchers?
- What mentorship model(s) do you adopt in mentored student research experiences in your own research group and how can it/they be transposed to the classroo

Research and Education together:

- CUREs require a wider vision of the role of the instructor than traditional classrooms because of the need for emotional and research support of students (Cooper et al., 2022; Goodwin et al., 2021; Linn et al., 2015; Shortlidge et al., 2016).
- Different models of the role(s) of instructors in research experiences have been developed, but the consensus is that mentorship covers five critical categories of skills: research, diversity/culture, interpersonal, psychosocial, and sponsorship (Gentile et al., 2017). These five branches of mentorship lead instructors to promote the development of adaptive attitudes in students.
- Instructors can encourage constructive attitudes through (1) framing and scaffolding (designing transparent tasks associated with an explicit awareness of both the pedagogical and research significance of the work), (2) interventions and pedagogical activities (promoting high student efficacy), and (3) explicit integrations of inclusive teaching practices (creating a supportive learning environment).
- Just like explicit expectations and relevance of CURE tasks need to be communicated to students, clear expectations and ELOs should be developed for the work of postdoctoral researchers, graduate student instructors, or peer undergraduate assistants who are engaged in the CURE. These goals should incorporate the personal and professional goals of the postdoctoral researchers, graduate student instructors, or peer undergraduate assistants (see Mabrouk, 2003).
- The TILT framework (see above) can be useful in framing both student activities and postdoctoral researchers, graduate student instructors, or peer undergraduate assistants’ tasks.
- Instructors should strike a balance between “being overly prescriptive, which inhibits creativity and agency, and being insufficiently supportive, which leads to uncertainty, frustration, and a sense of failure” (Hanauer et al., 2012:384).
- Many activities and learning exercises are helpful in engaging students in experiences that promote student efficacy, including:
  - Sharing their personal goals for the CURE and expressing their expectations for the CURE.
  - Participating in the social network of their research team through meetings, discussion boards, etc.
  - Explicitly associating elements of the research process with valuable skills and attitudes for their career aspirations.
  - Articulating hypotheses and questions in the context of the CURE’s research goals.
  - Conducting experiments, collecting and organizing data.
  - Analyzing and interpreting data. Evaluating evidence. Critiquing conclusions.

- Becoming aware of the necessity of experimental failure.
  - Understanding that sometimes discoveries emerge from iterative processes.
  - Considering the quality of evidence and their relevance to the argument.
  - Synthesizing results and drawing conclusions; planning next steps.
  - Reading the primary literature, attending relevant seminars, and discussing their work with others.
  - Presenting progress reports and comparing ideas in group setting.
  - Reflecting on how the process of critique contributes to research progress.
- Consider guiding the cognitive exploration of students by implementing these recommendations modified from Cooper et al. (2022):
  - Challenging students to check if their hypothesis is explanatory, clearly stated, and distinguishes between multiple ideas.
  - Pushing students to fully explain ideas by asking follow-up questions and asking for clarification.
  - Highlighting potential pitfalls of hypotheses and protocols encouraging student to identify the problem and its solution.
  - Asking students to explicitly articulate links between research goals, hypothesis, and experiment.
  - Encouraging students to consider alternative experimental outcomes or explanations for their predictions or results.
  - Redirecting student ideas that are unproductive by bringing student attention back to their original hypothesis/goal.
  - Reminding students to think about controls they need to consider in their experiments/analyses.
  - Providing assistance with analytical tools or protocols that are merely means to an end and not critical elements of the learning goals, to enable students to focus on course objectives.
  - Assessing the time management of the students and propose adjustments to protocol or timeline accordingly.
- Instructors can foster a supportive environment by integrating best practices developed from student focus groups (Faulkner et al., 2021) including the following:
  - Contacting students: consider emailing or connecting with students before the semester begins. This initial contact is not only meant to offer practical information, such as class time and location, it also opens the door for a connection between students and instructor. Activate your class website before the class begins. This allows you to post messages for students and gives them the opportunity to reach out with questions.
  - Learning about students: you can promote inclusivity by learning about the identities, circumstances, and concerns of your students. You can do so through get-to-know-you surveys as well as start-of-the-semester conferences. This is a good time and place to ask students their pronouns, names, and the pronunciation of the latter.
  - Setting the right tone for the class: the first day of class sets the tone for the rest of the semester. As such, it is a critical time to establish an inclusive environment. Creating an inclusive classroom necessitates that each student feels like you see them as a unique individual. It also requires the classroom environment to reflect respect and care for all students. You should set expectations for class discussions and respect of everyone’s background. You also need to encourage everyone to express their ideas and learn from each other. Communicate that you will not tolerate any discriminatory attitudes or behaviors in the classroom.
  - Encouraging introductions and setting group atmosphere: creating a group atmosphere that cultivates student collaboration and support throughout the semester is essential for inclusive pedagogy. It helps students create relationships with other students in the classroom and encourage collaboration and support. You can facilitate this collaboration by having students introduce themselves and exchange contact information with the people they will work with.
  - Explaining syllabus and expectations: transparency is critical to a supportive class environment. Going over the CURE in detail on the first day of class is a useful way to make students feel welcome and helps you establish course expectations.
  - Self-disclosure: the classroom can feel more inclusive when you share aspects from your own life. In particular, it may help students feel less vulnerable. Students see self-disclosure both as a way to get to know you and a sign of mutual respect.
  - Being approachable: see your students as the people they are by using their name and correct pronouns, treat them as capable learners, respect them, encourage them, check in on them, make them feel welcome, and do not reinforce rigid power hierarchies. Explicitly tell students how you prioritize their learning and needs as well as your desire to help and support them.
  - Staying engaged: pay attention to both the verbal and nonverbal responses of students. Notice the silences and apathy as much as the participation. Hold office hours and consider requiring students to set up a five-minute meeting several times throughout the semester. Holding hybrid office hours both in-person and through online tools like Zoom or Teams can facilitate attendance.
  - Providing resources: give students information about campus and community resources, not only on the first day of class, but also throughout the semester. Provide specific directions for resources including academic services, clubs, tutoring, special interest groups, as well as mental health and wellness support.
  - Encourage reflection: reflect on your experience in the course, how you engage students in dialogue, how you keep communications with students open, and how you include everyone in the class discussions. Examine your own positions of power, privilege, and vulnerabilities. Encourage your students to do the same and reflect upon their roles in the classroom environment and their interactions with others.
- Consider also the guidelines from Cramer and Prentice-Dunn (2007) for an alternative framing of the elements above into a mentorship model “caring for the whole person”.
- There is an important role for peer-mentoring among students engaged in the CURE. Such mentoring mimics the interactions between researchers and, along with mentorship from more experienced researchers (instructor of record, postdoctoral researchers, graduate teaching assistant, etc.), can help mitigate the frustration inherent to the failures and setbacks of authentic research experiences (Hanauer et al., 2012).
- The expert-novice divide is well recognized across disciplines (Inglis & Alcock, 2012; National Research Council, 2000; Newman et al., 2021; Stofer, 2016) and can sometimes represent an obstacle to mentorship; it may be overcome by involving peer teaching assistants. Undergraduates with prior experience of the CURE or mentored research experience outside of the classroom may be perceived as more approachable than more senior instructor(s); they also have recent experience with the struggles and challenges associated with the knowledge and skills taught in the CURE. They can provide very valuable feedback, including on writing assignments (Cho et al., 2006). Research shows that undergraduate peer assistants are valued by students enrolled in CUREs and facilitate their success (Olson et al., 2022).
- Although peer assistants can sometimes struggle with their identity and role in the classroom (Terrion & Leonard, 2007), training and mentorship of peer assistants can help overcome these difficulties (Handelsman et al., 2005).
- Requiring members of the instructional team to schedule dedicated time to mentoring in their week leads to more successful experiences (Shanahan et al., 2015; Terrion & Leonard, 2007). This is because time scarcity is a major barrier to effective research mentorship (Gentile et al., 2017).
- Communication with postdoctoral researchers, graduate student instructors, or peer undergraduate assistants around the support they need and the resources that would enhance their experience (in both content and format) is important in creating a successful teaching experience for them (BrckaLorenz et al., 2020).
- Goodwin et al. (2021) identified different mentorship roles for CURE graduate student instructors and provided evidence that instructors who embrace their functions as “student supporters” and “research mentors” are more likely to see value for themselves in the CURE and engage in teaching CUREs again. Goodwin et al. (2021) defined “student supporters” as mentors who “[provide] emotional support to students” and “research mentors” as instructors who “[develop] student[s’] autonomy and competence as researcher[s]” (Goodwin et al., 2021: Fig. 3).
- Good mentorship should incorporate socioemotional support, culturally relevant mentoring, and appropriate personal interest in students (Haeger & Fresquez, 2016; Robnett et al., 2018; Shanahan et al., 2015). Consider also sharing your own stories of struggles and failures with research (Jayabalan et al., 2021).
- The affective-motivational research competence model (Wessels et al., 2018) has identified six situations that mentees need support with to engage in research as well as dispositions that can be fostered to help overcome the challenges of these situations (**Table 16**). These dispositions can be brought about through interventions and scaffolding activities of the CURE aimed at fostering specific experiences and tasks that promote knowledge integration (Linn et al., 2015).

Situations	Dispositions	Experiences
Developing a research interest	Curiosity and thirst for knowledge Value-related interest in research Finding joy in conducting research	➢ Share goals for the CURE relative to personal and career aspirations ➢ Express expectations for research experience ➢ Participate in social network of research team
Making decisions	Tolerance for uncertainty Acceptance of narrowing down	➢ Articulate hypotheses and questions about the research topic ➢ Identify or formulate a question in the context of the CURE's research goals ➢ Conducts experiments, collect, and organize data ➢ Analyze and interpret data. Evaluate evidence. Critique conclusions
Enduring setbacks	Tolerance for frustration	➢ Experience experimental failure ➢ Consider how discoveries emerge from iterative processes ➢ Recognize strengths related to career aspirations
Unravelling irritations	Tolerance for complexity	➢ Consider quality of evidence and relevance to argument ➢ Synthesize experimental results ➢ Make final conclusions and plan next steps
Making use of feedback and critiques	Willingness to seek and accept feedback	➢ Read literature, attend seminars, discuss with research team ➢ Experience how process of critique contributes to research progress; share ideas as a team
Audience-appropriate communication of research	Acceptance of divergent perspectives	➢ Read literature, attend seminars, discuss with research team ➢ Present progress reports and compare ideas in group setting

Table 16. Experiences and tasks that mentors should use to foster adaptive dispositions in students that help them through (modified from Linn et al., 2015) challenging situations (modified from Wessels et al., 2018).

Box 10: Important points

- A CURE is an opportunity to move away from a model of teacher-identity centered around the delivery and assessment of knowledge to one of a mentor.
- Successful CUREs empower students to think for themselves and enable them to make mistakes in a safe and supportive environment.
- A CURE is an authentic research experience and as such can introduce students to collaboration by including outside researchers, laboratory personnel, and staff members.
- Every member of the CURE has goals, needs, and expectations. This is true of the students and instructor(s) of course, but also applies to any other member of the team. Consider carefully the help and means everyone requires to achieve their mission.

11. How will the success of the CURE be assessed?

Overview: Just like the assessment of the students' work in the CURE enables them to reflect upon their progress and make corrections, the assessment of the CURE itself provides opportunities for redesigns, corrections, and reflection by the instructor(s). Assessing students' work provides a basis for an often-necessary grade. Similarly, the evaluation of the CURE is integral to the instructor's annual evaluations, their eligibility for promotions, tenure, and awards. Because CUREs often require financial and/or logistical support, the assessment of the experience enables the demonstration of its efficacy and can facilitate the renewal or expansion of the program. When a CURE is an integral component of the curriculum, particularly as an element of introductory courses, general education requirements, or program prerequisite, its evaluation is an important part of the overall educational experience of students that may become part of program reviews, external evaluations, and validation of the course as satisfying specific certifications or endorsements. There are three overlapping components to the assessment of CUREs: course, instructor, and student outcomes (Brownell & Kloser, 2015) summarized in **Figure 4**. Section F presents a discussion of the existing approaches in evaluating all three of these elements of CUREs from a programmatic and scholarly point of view; many of these approaches are aimed at incorporating data from large number of students; several may require the involvement of education research collaborators or institutional representatives. Here, the focus is on data collection and analyses that individual instructors can engage in to reflect upon the research progress and the pedagogical framework.

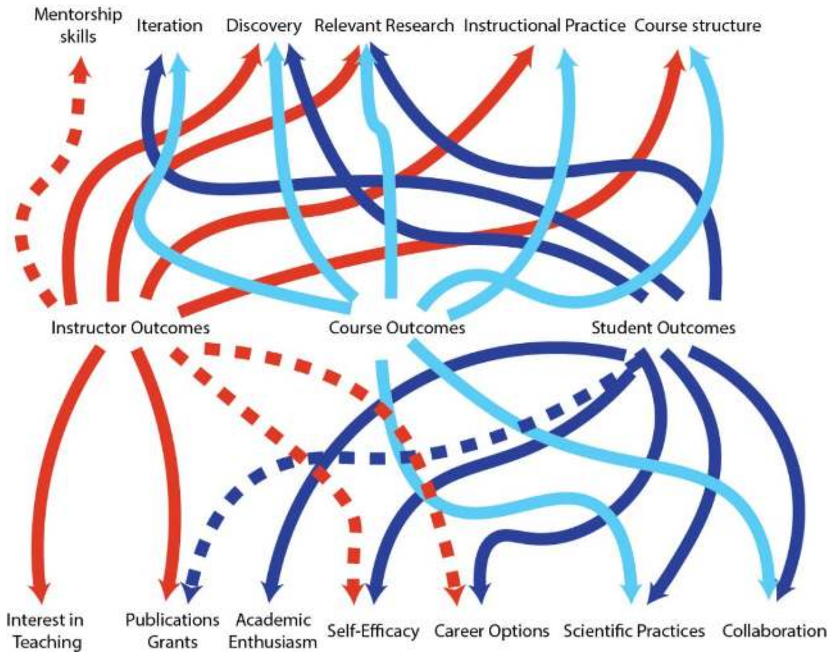

Figure 4. CURE assessment "meta-framework" (redrawn and modified from Brownell & Kloser, 2015). Dashed lines were added to reflect the importance and value of outputs (publications and grants) to students, and the roles of graduate students, peer-assistants, and postdoctoral researchers as instructors.

This section enables you to determine:

1. The value of the individual assignments and activities implemented
2. The usefulness of the scaffold
3. The fit of the assessment and learning goals
4. The research outcomes of the CURE and their significance to stakeholders
5. The success of the CURE in promoting the success of all students

Education starts here:

- Are the assessments aligned with the learning goals of the course?
- Are the formative and summative assessments of the students' learning also assessed for the success of the course activities and scaffold?
- Does the scaffold for the CURE need to be revised?
  - Are there bottlenecks to learning that remain to be addressed by the CURE scaffold?
  - Are some parts of the course "over-scaffolded"? Is the preparation of the students underestimated for specific activities?
- Are there concept inventories or professional society standards that can be used to assess students' proficiency on certain topics of the CURE?
- Did students encounter problems with specific activities?
  - Are there assignments that were unclear?
  - Were there homework assignments that multiple students asked for clarifications about?
- What are the perceived gains from the CURE that students express?
- Can specific assignments' rubrics be edited to include elements useful to assessing the efficacy of the activity?
- Does your institution have survey tools (including some incorporated into student evaluations) that can be leveraged to assess your pedagogy of the course?
- Can you engage peer instructors in reviewing your course?
- Can you present the CURE you developed at an education conference or in the education session of the annual meeting of your professional society to get feedback and thoughts from colleagues?
- Can you gather data on the success of the CURE in closing the equity gap and promoting the learning and success of all?

Scholarship starts here:

- Can the research findings of the CURE be presented to the research community for feedback?
  - Is the work ready to be submitted in the form of a proposal or manuscript?
  - Is it appropriate to present the work at a conference?
- Are there other stakeholders of the research who could provide critiques of the work and assess its significance?

Research and Education together:

- The efficacy of the scaffold and activities can be in part assessed through the performance and success of students. Each assessment of the students' work is also an assessment of the work of the instructor. In many ways, some of the issues associated with CURE assessment can be answered through student assessment, an issue discussed earlier in the document.
- A key element of the assessment of the activities and assignments of the CURE is to define clear goals for each of them. This participates in increasing transparency for students and enables the determination of their success and therefore yours.
- Assessments of students' work and survey tools can be used to identify knowledge/skill bottlenecks to address in iterations of the CURE.
- There are numerous forms of formative assessment that can be implemented throughout the CURE to help determine the efficacy of the CURE activities and interventions:
  - Minute papers or muddy point papers (Anderson & Burns, 2013; Stead, 2005) can help assess whether or not a particular activity fulfilled its learning goals. Example of prompts are available from multiple sources (e.g., Tufts University; University of Wisconsin)
  - Concept maps and Venn diagrams can help students synthesize information and be assessed by instructor(s) for learning and efficacy of scaffolding activities (Bauman, 2018; McConnell et al., 2003).
  - Group and individual presentations of research updates enables the instructor to assess the efficacy of the scaffold in supporting the research progress.
- Group reviews and self-reviews ([**Activity 19**]) can help assess group dynamics and the effect of team building efforts.
- Questions can be included in the reflection workbook ([**Activity 26**]) to enable the assessment of the students' understanding of concepts presented in the CURE activities, thus determining their efficacy. Student reflections can be evaluated with students to get additional insights in the student experience (McLean et al., 2022).
- Many institutions allow the design of a few custom questions to add to student evaluations of instruction. These custom questions can be used to assess the pedagogy of the course. Examples of questions can be found in the project ownership survey (Hanauer & Dolan, 2014), the classroom and school community inventory (Rovai et al., 2004), the laboratory course assessment survey

(Corwin et al., 2015), the classroom undergraduate research experience survey (Denofrio et al., 2007), the science process skills inventory (Arnold et al., 2013), or a combination of these tools (Lo & Le, 2021).
- Some surveys have been developed to specifically assess the efficacy of CUREs, or even specific activities of the course (e.g., Satusky et al., 2022)
- Online survey tools (e.g., Qualtrics and SurveyMonkey) or paper survey tools (including some standard student evaluation forms) administered as part of midterm evaluations and end-of-term evaluations can enable instructors to poll students on affect towards specific activities, perceptions of learning gains, and self-efficacy.
- Concept inventories (**Table 6**) as well as professional standards (e.g., American Psychological Association) can be used to devise summative assessments.
- Many indicators of learning outcome satisfaction can be assessed by integrating appropriate criteria into grading rubrics (Chamely-Wiik et al., 2014).
- Extensive notetaking of class observations, student behaviors, reflections on pedagogy, problems encountered in the classroom, failures, and successes can be used to revise individual activities as well as their organization.
- Interviews and focus-groups by the instructor or by a neutral third party can enable students to share their thoughts and feelings about the CURE and its outcomes (e.g., (Brownell & Kloser, 2015; Turner et al., 2021; Wooten et al., 2018). This approach can also be used to assess the experience of graduate student instructors and peer assistants to ensure positive and improved experience for all members of the CURE team (see Heim & Holt, 2019). Essays can also be used to gather the thoughts of CURE participants (Wooten et al., 2018).
- Partnering with colleagues and education researchers can be helpful in determining the impact of the CURE on all students and the success of the CURE in mitigating systemic biases and enabling the learning of a diverse student body.

Box 11: Important points

- Take abundant notes during the implementation of the CURE to guide redesigns, expansions, and modifications to the course
- Ask yourself numerous questions when grading and/or assessing student work, for example:
- Did this assignment fulfill its goal of assessing the associated ELOs?
- Did this activity support the associated learning goal(s)?
- Does your institution enable custom questions on student evaluations?
- Can you use surveys or include questions within assignments to gather data from your students on the activities and assignments of the CURE?
- Reviews by peers of manuscripts, conference presentations, and grant proposals are a measure of the success and progress of the research of the CURE